

Lower limb nerve anatomy & block technique

*Aduqna A.
May, 2024*

Outline

- Introduction
- Lower limb nerve Anatomy
- Femoral nerve block
- Lateral cutaneous nerve of the thigh block
- 3 in 1 block
- Sciatic nerve block
- Popliteal nerve block
- Ankle block
- References

Introduction

- Lower extremity PNBs have NEVER been as widely taught or used as other forms of regional anesthesia.
 - a) Entire lower extremity cannot be anesthetized with a single injection,
 - b) Injections are generally deeper than upper extremity block.
 - c) Neuraxial techniques are widely taught and used alternative methods for providing reliable lower-extremity anesthesia.

Advantages of Lower extremity PNB

- On Arrival block
- Preemptive analgesia
- Post operative pain relief
- Rescuing a risky patients
- Less complications

Lower Limb innervations

- The innervations of the leg is derived from the lumbar and sacral plexuses
- Lumbar plexus formed by L1–L4

The main branches are

1. Lateral cutaneous N of thigh
2. Femoral N
3. Genitofemoral N
4. Obturator N

Lower Limb innervations...

- Lumbo sacral plexus formed by L4 –S5
- Main branches are
 1. Sciatic nerve
 2. Posterior cutaneous nerve of thigh
 3. Pudendal nerve

Lumbo sacral Plexus



- Lateral cutaneous N of thigh

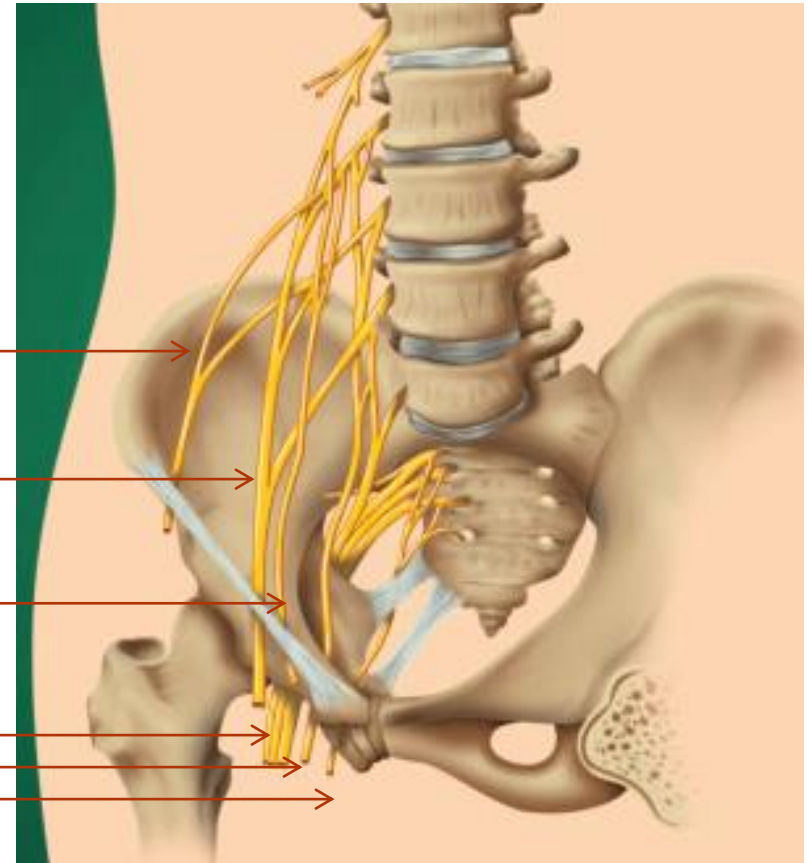
- Femoral N

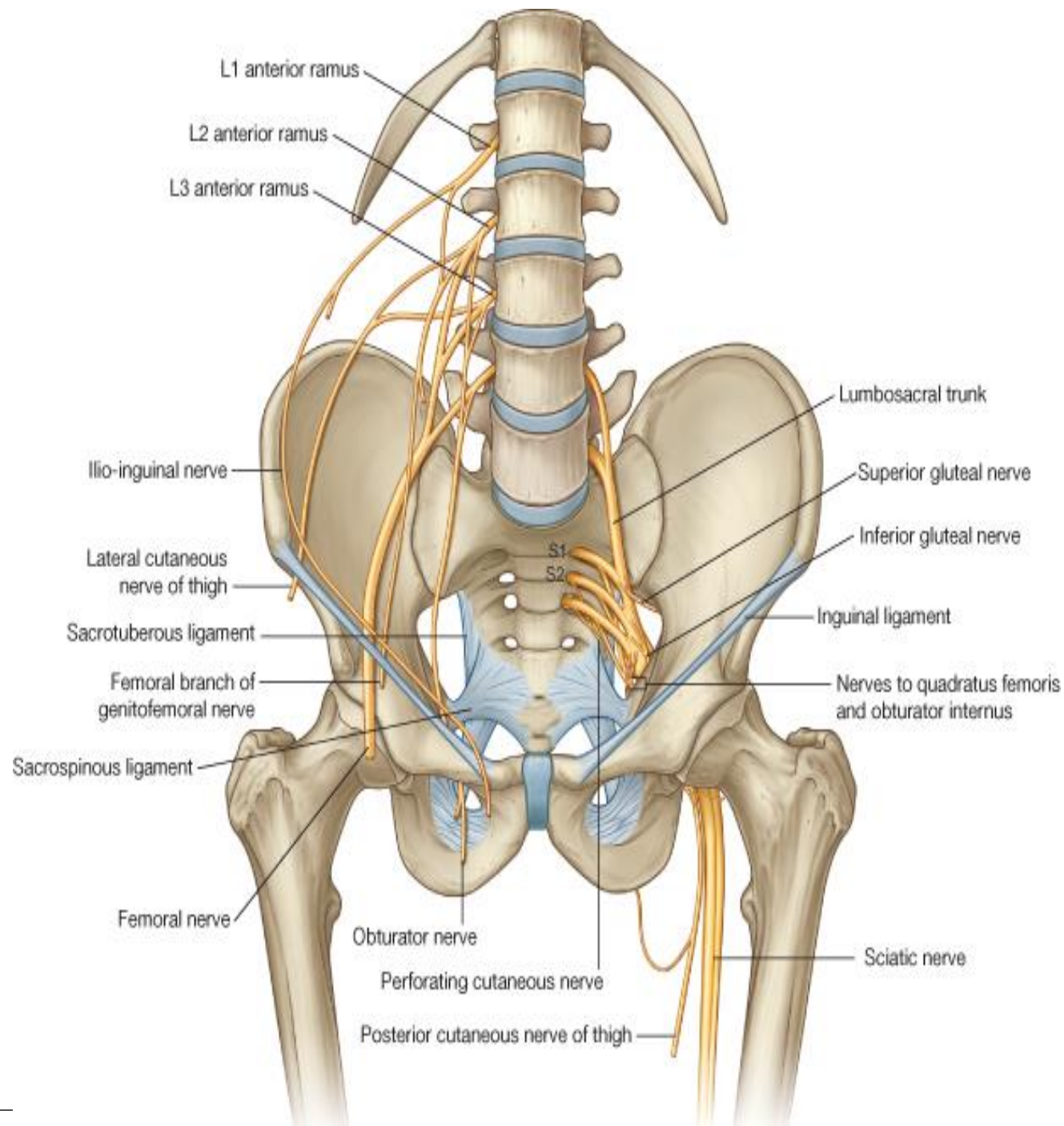
- Genitofemoral N

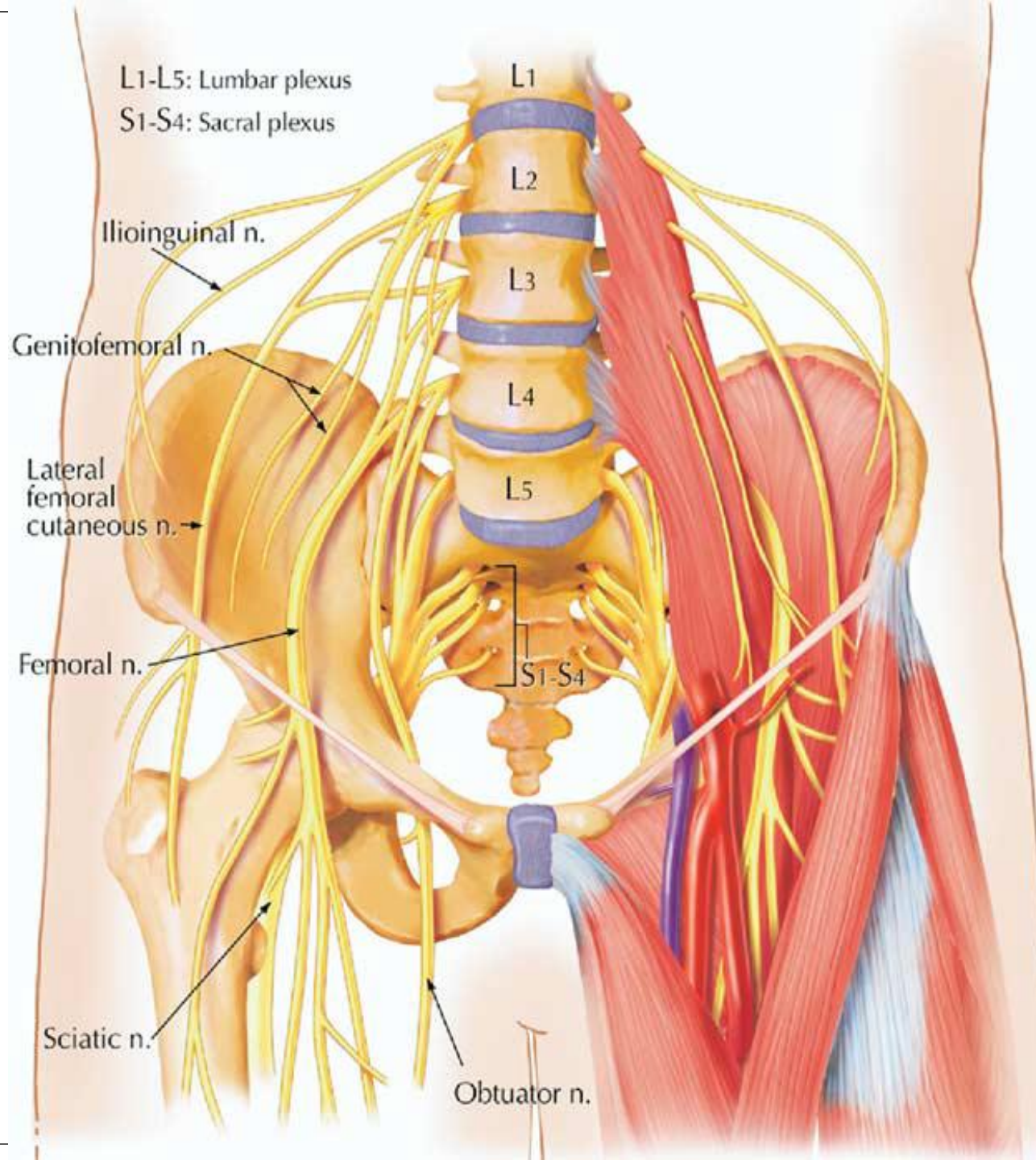
- Sciatic N

- Obturator N

- Pudendal N

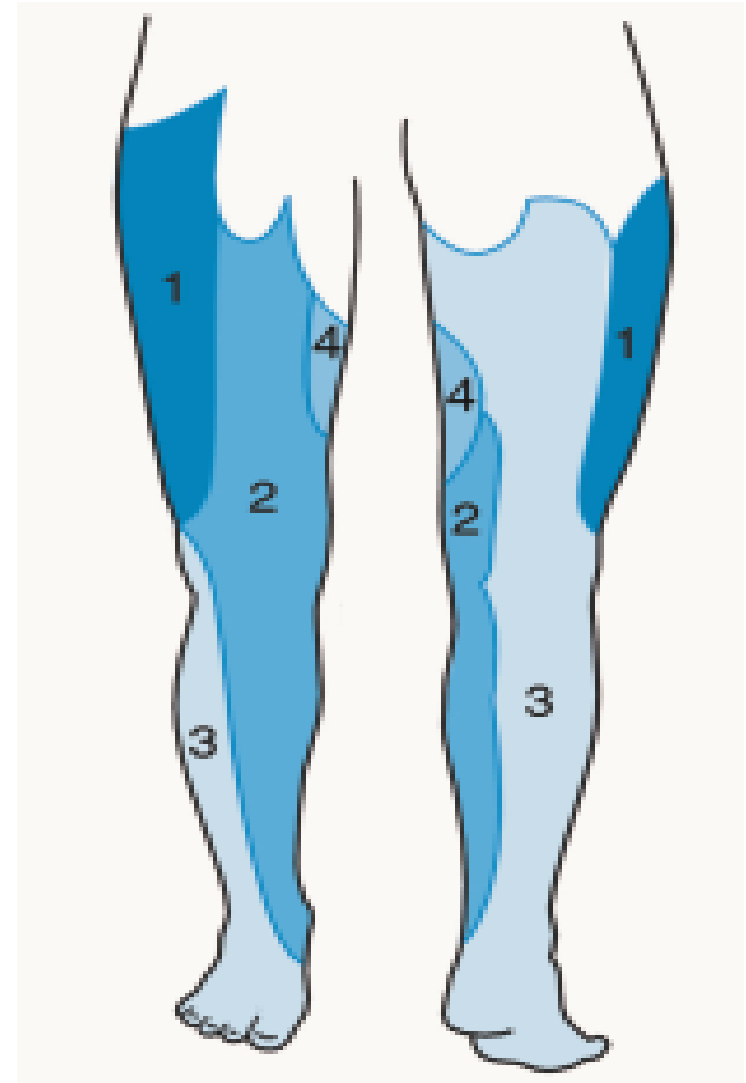






Distribution of Lumbo sacral plexus

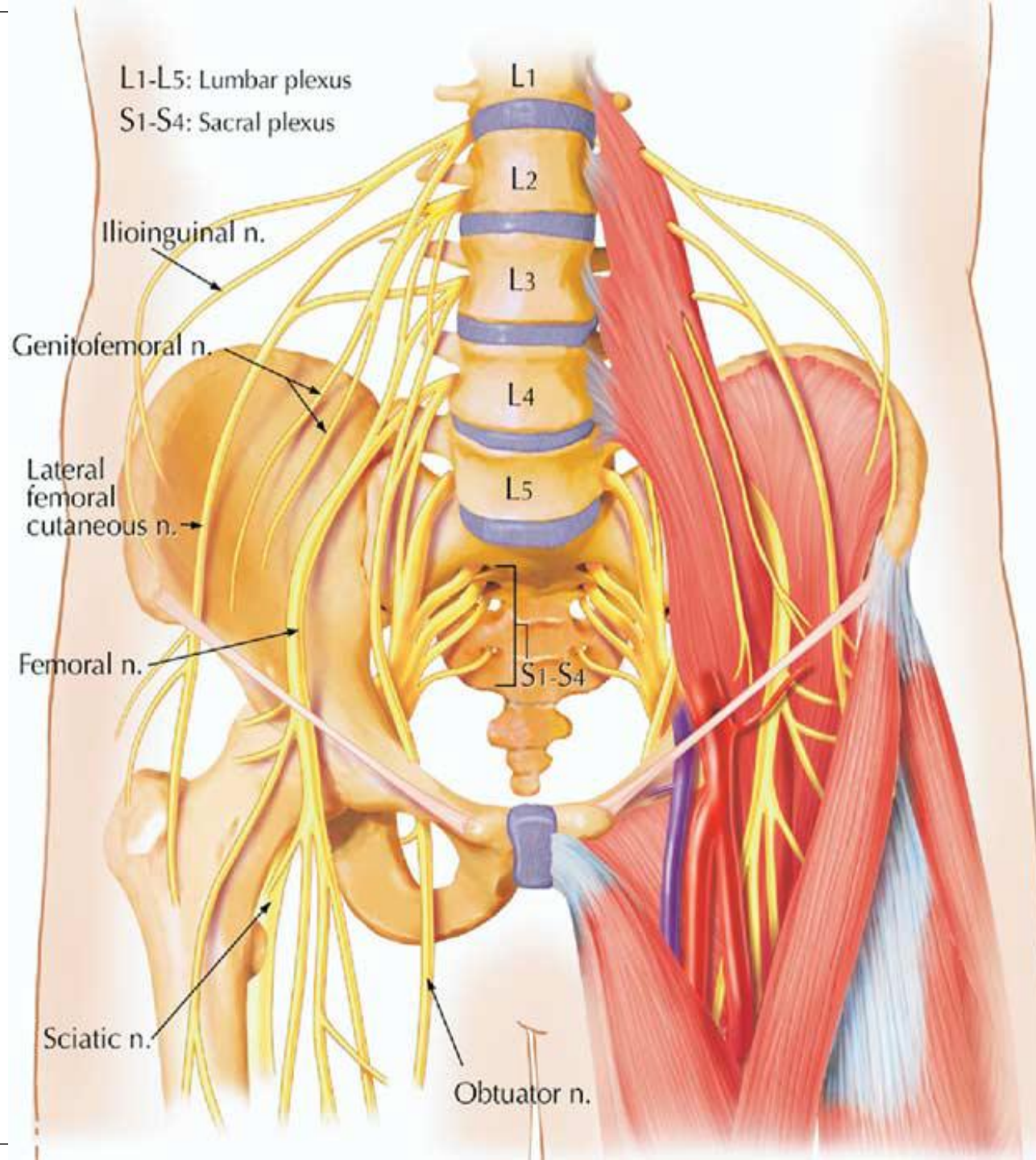
1. Lateral cutaneous nerve of thigh
2. Femoral nerve
3. Sciatic nerve
4. Obturator nerve



Lumbar plexus Anatomy

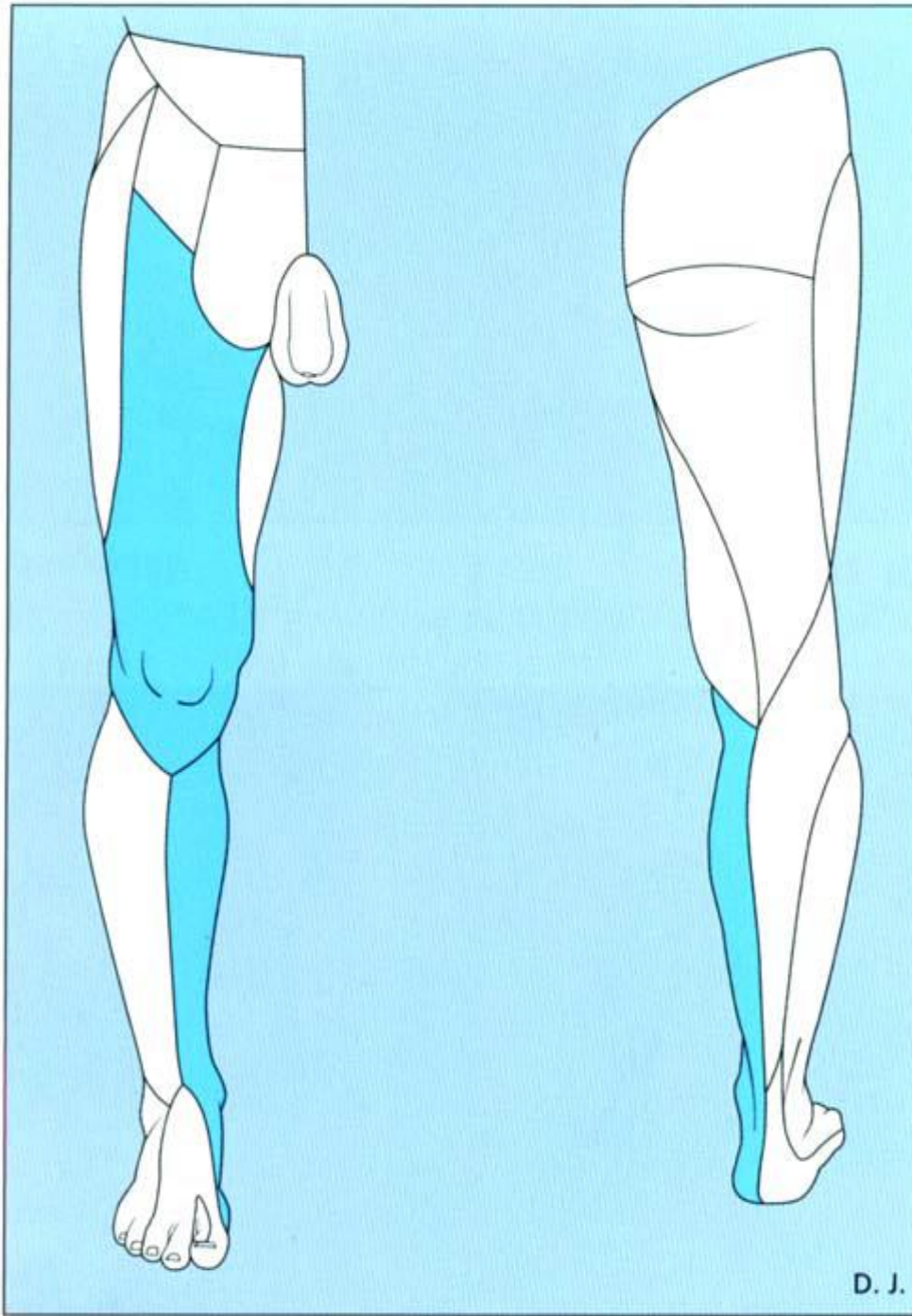
Femoral Nerve

- Formed by the 2nd, 3rd, and 4th lumbar nerves.
- It enters the thigh posterior to the inguinal ligament.
- There it lies lateral and posterior to the femoral artery.
- This relationship to the femoral artery exists near the inguinal ligament, but not after the nerve enters the thigh.



Femoral cont'd...

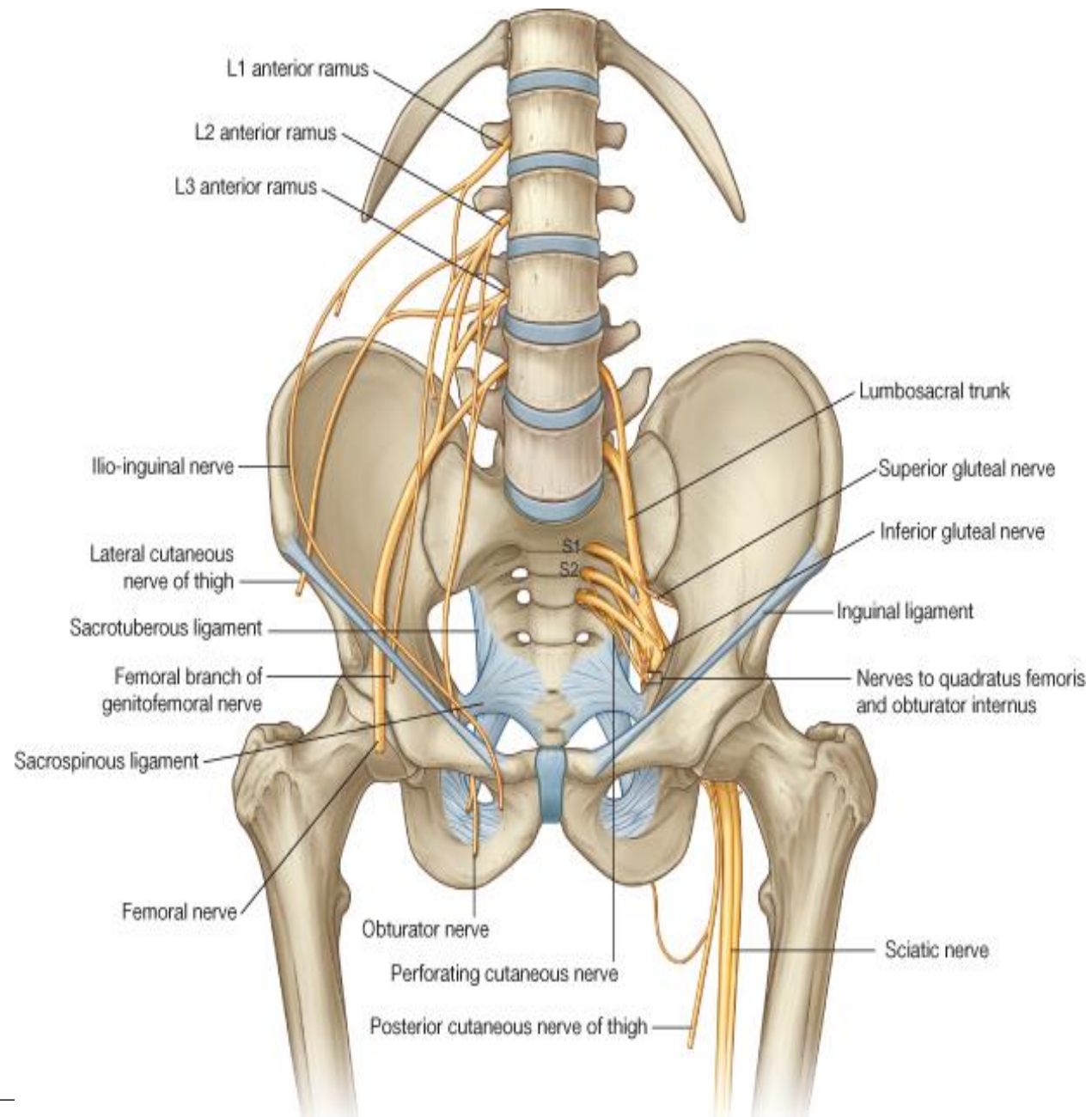
- Generally the femoral nerve innervates:
 - All muscles in the anterior compartment of the thigh;
 - Skin over the anterior aspect of the thigh, anteromedial side of the knee, the medial side of the leg, and the medial side of the foot.



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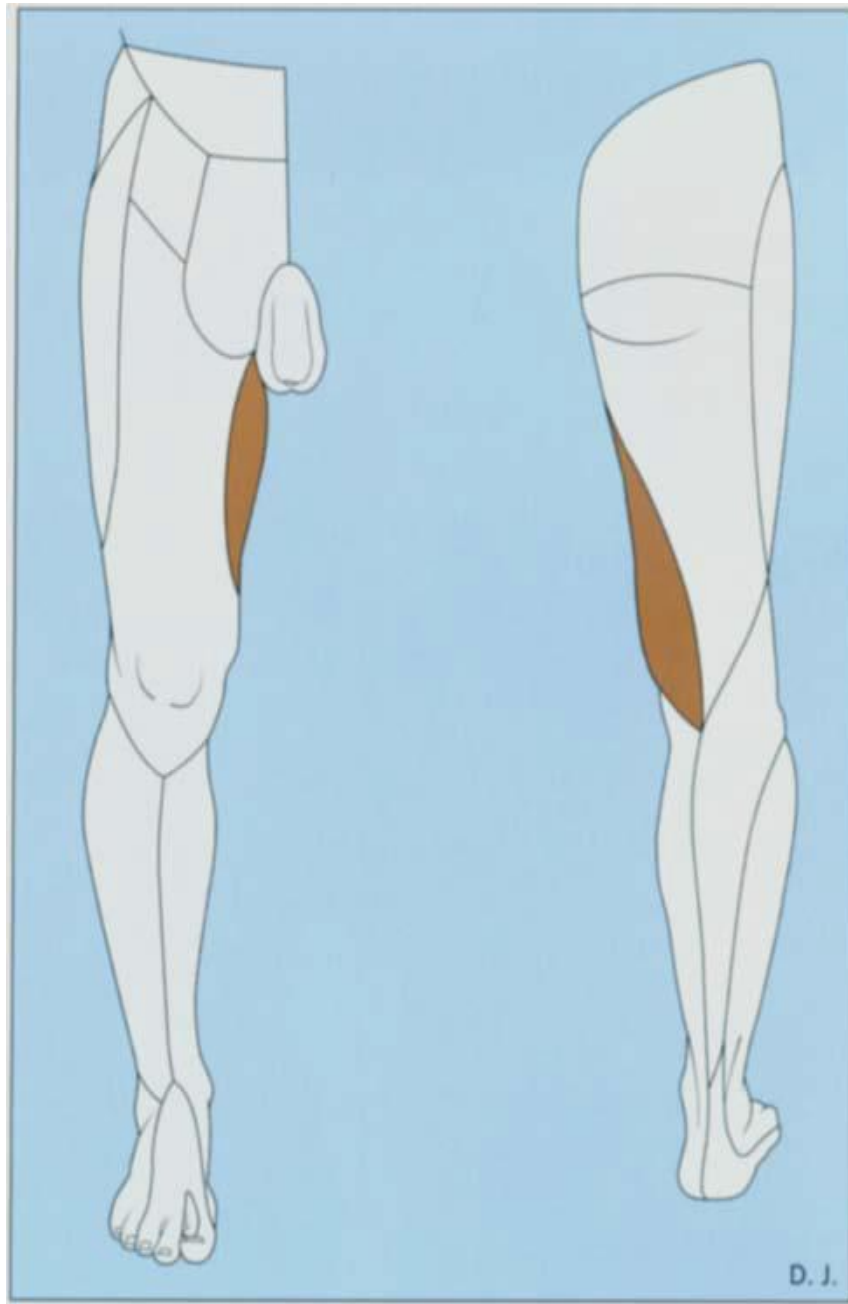
Obturator Nerve

- Like the femoral nerve, originates from L2 to L4.
- It descends along the posterior abdominal wall, passes through the pelvic cavity and enters the thigh by passing through the obturator canal.
- After entering the thigh through the obturator groove, the nerve divides into an anterior and posterior division.



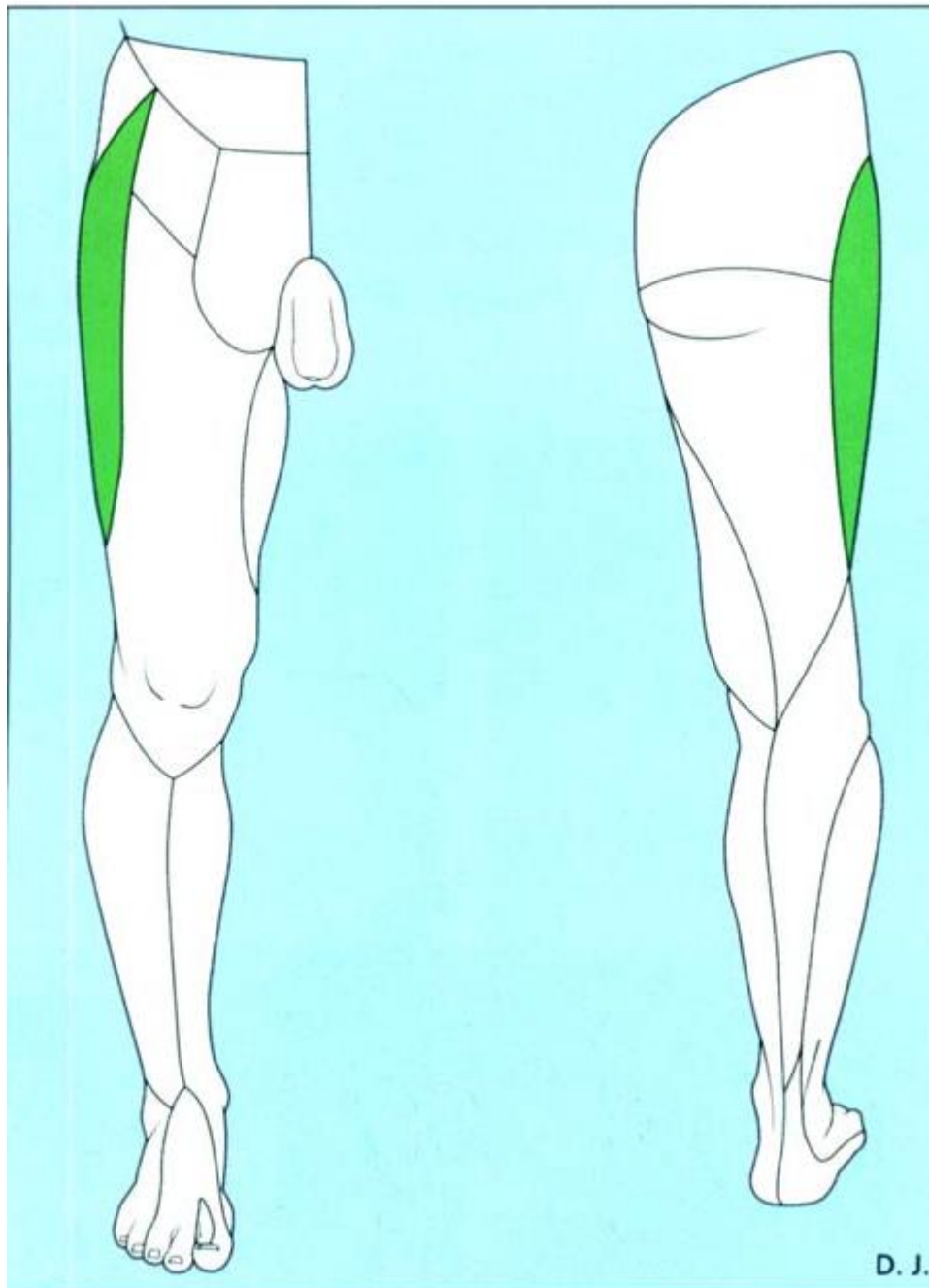
Obturator cont'd...

- At the level of the inguinal ligament, the obturator nerve lies **deep and medial** relative to the femoral nerve and is separated from it by **several fascial compartments**.
- The obturator nerve innervates:
 - All muscles in the medial compartment of the thigh
 - The obturator externus muscle;
 - Skin on the medial side of the upper thigh.



Lateral cutaneous nerve of the thigh

- Originates from L2 and L3.
- It leaves the abdomen by either passing through the gap **between the inguinal ligament and the pelvic bone** just medial to the anterior superior iliac spine or by passing directly **through the inguinal ligament**.
- It supplies skin over a widely variable aspect of the lateral side of the thigh.
- It has **no motor** innervation.



Cutaneous
innervation area of
the lateral femoral
cutaneous nerve

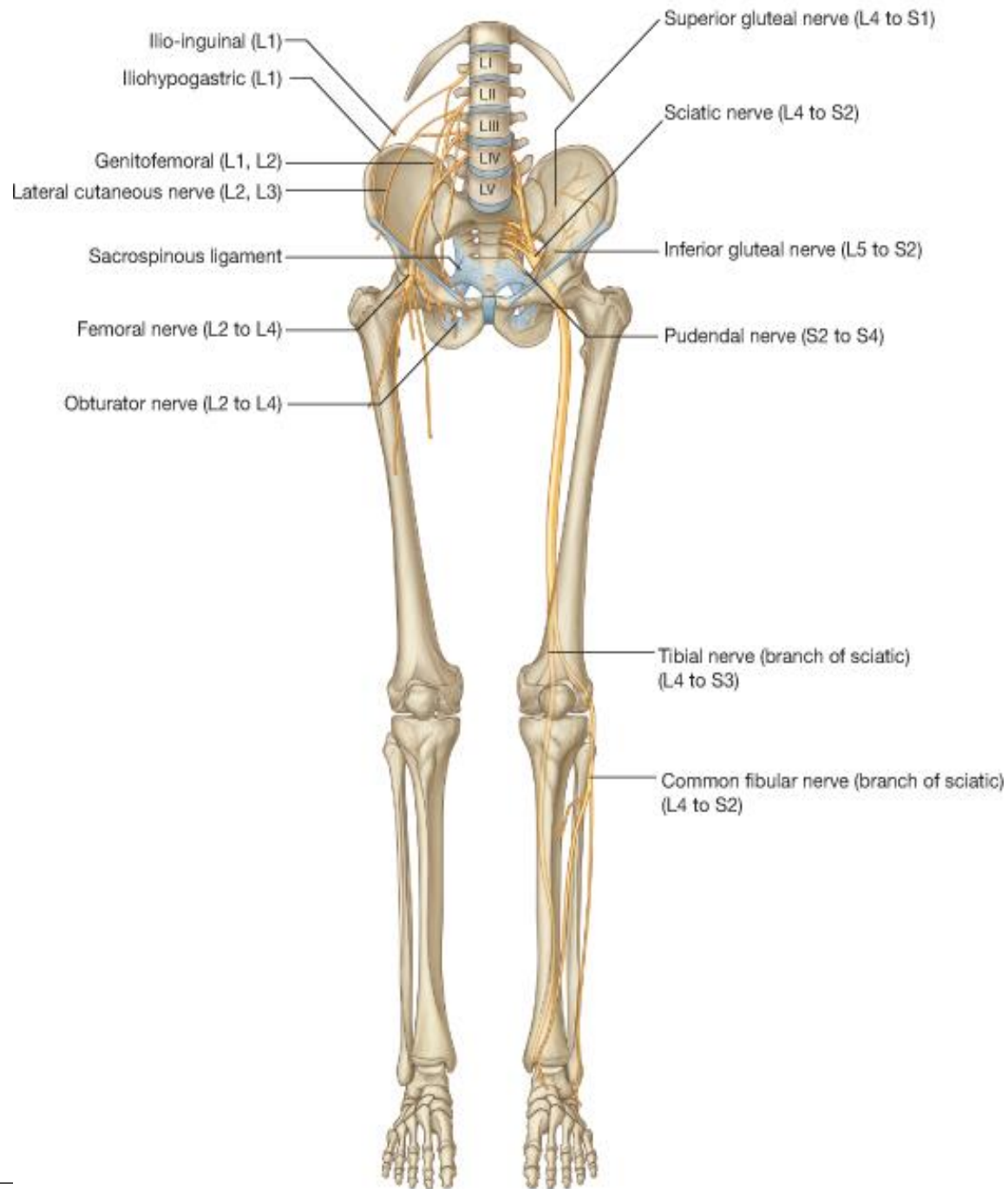
Sacral plexus Anatomy

Sciatic nerve

- Is the largest nerve of the body and carries contributions from L4 to S3.
- It leaves the pelvis through the **greater sciatic foramen** and passes through the gluteal region, and then enters the posterior compartment of the thigh where it divides into its two major branches:
 - The common fibular nerve;
 - The tibial nerve.

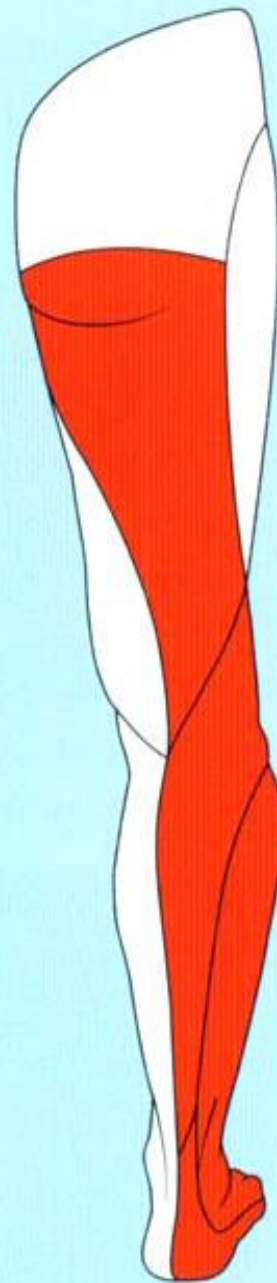
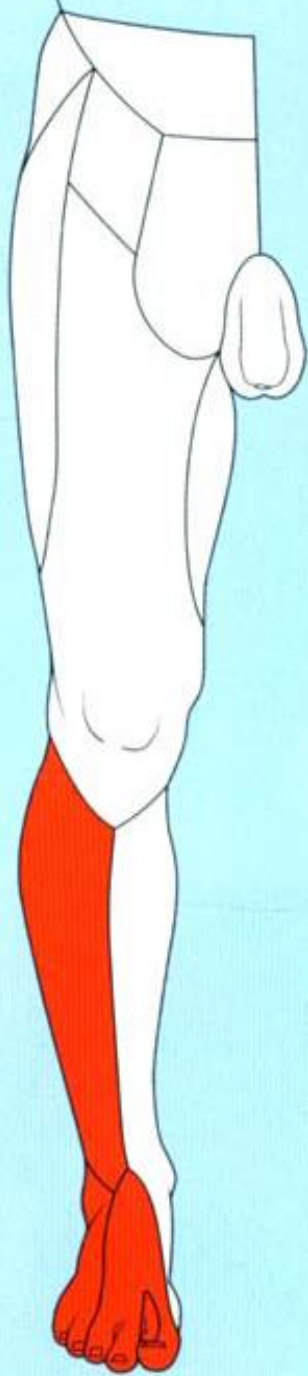
Lumbar

Sacral

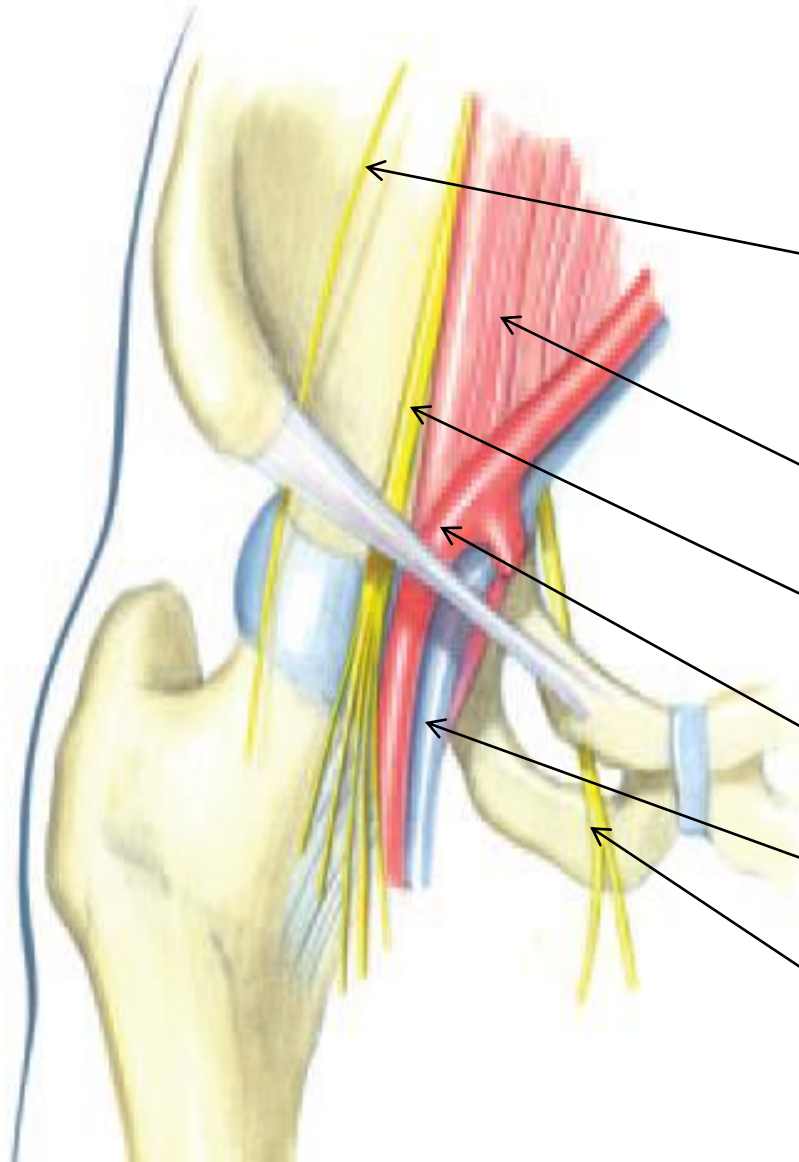


Sciatic cont'd...

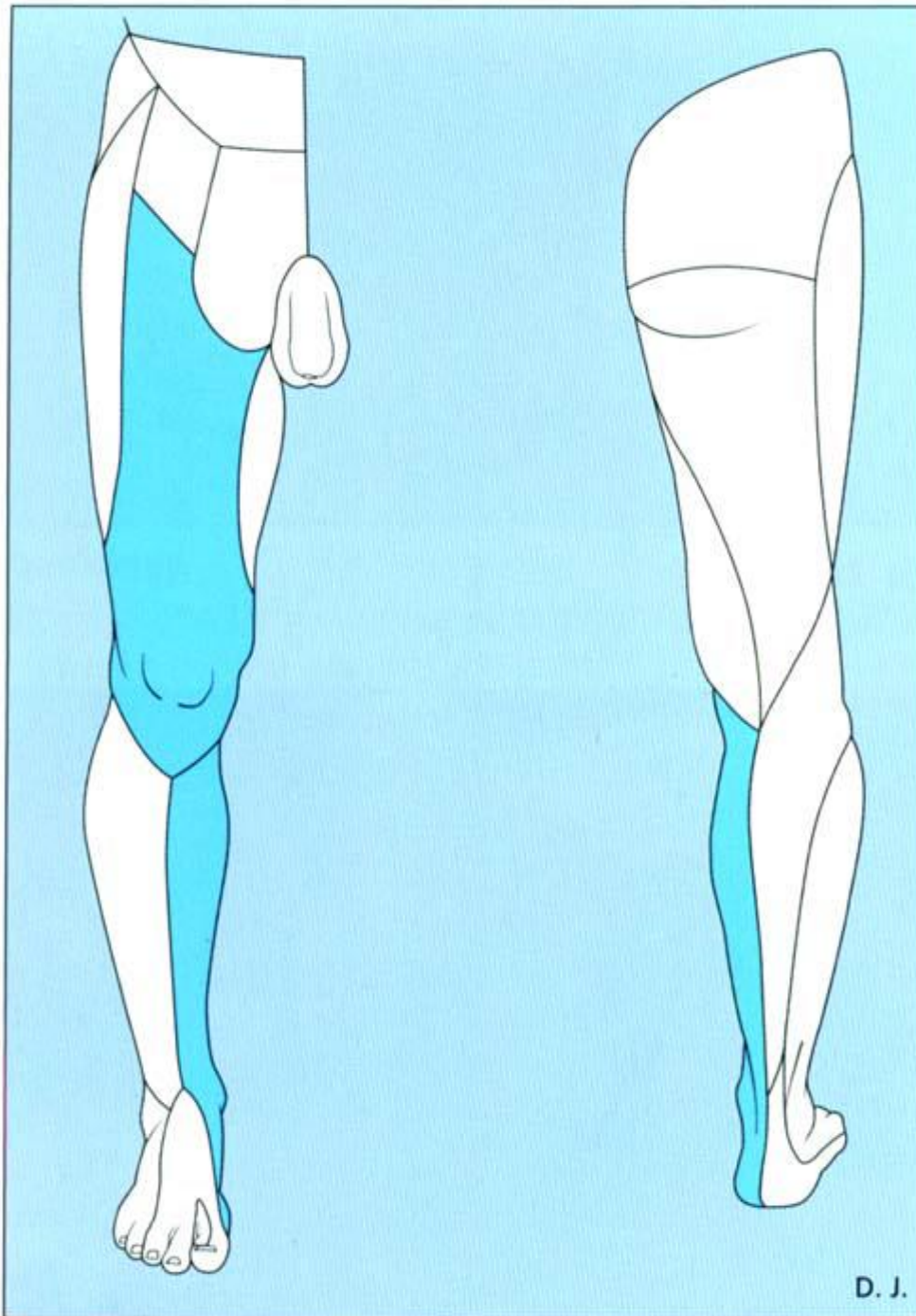
- The sciatic nerve innervates:
 - All muscles in the posterior compartment of the thigh;
 - All muscles in the leg and foot;
 - Skin on the lateral side of the leg and the lateral side and sole of the foot.



1. Femoral Nerve block



1. Lateral femoral cutaneous nerve
2. Psoas major muscle
3. Femoral nerve
4. Femoral artery
5. Femoral Vein
6. Obturator nerve



- Cutaneous innervation area of the femoral nerve

- Indication

- Solely – analgesia for fractured femoral shaft, postop analgesia after knee surgery
- Combination with sciatic – for knee and lower leg surgery with or without GA

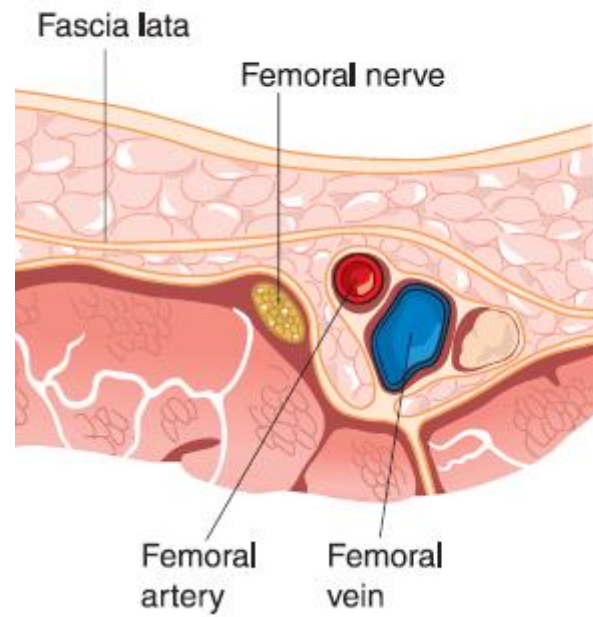
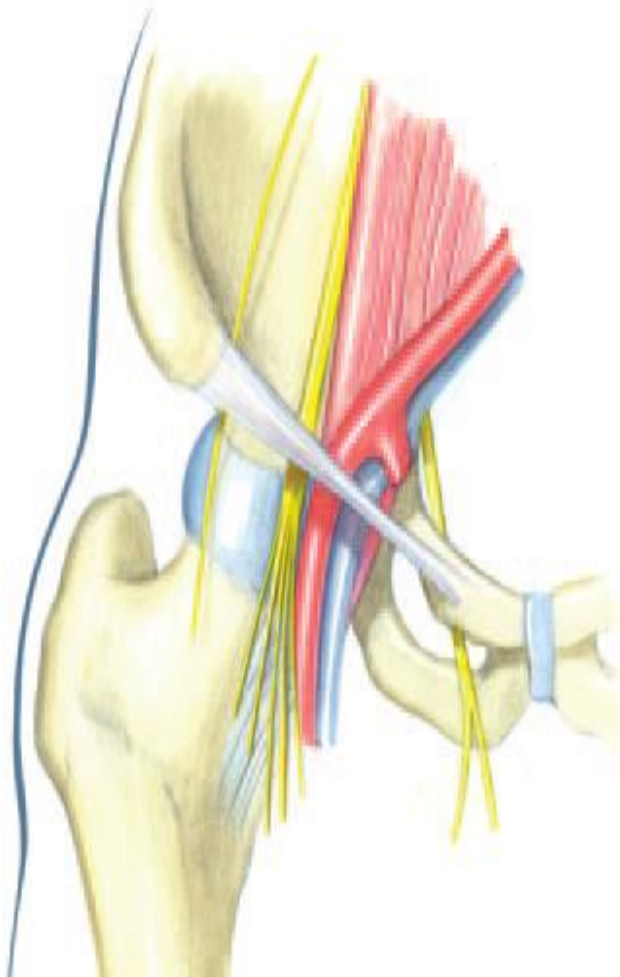
- Landmarks

- Inguinal ligament, Femoral artery

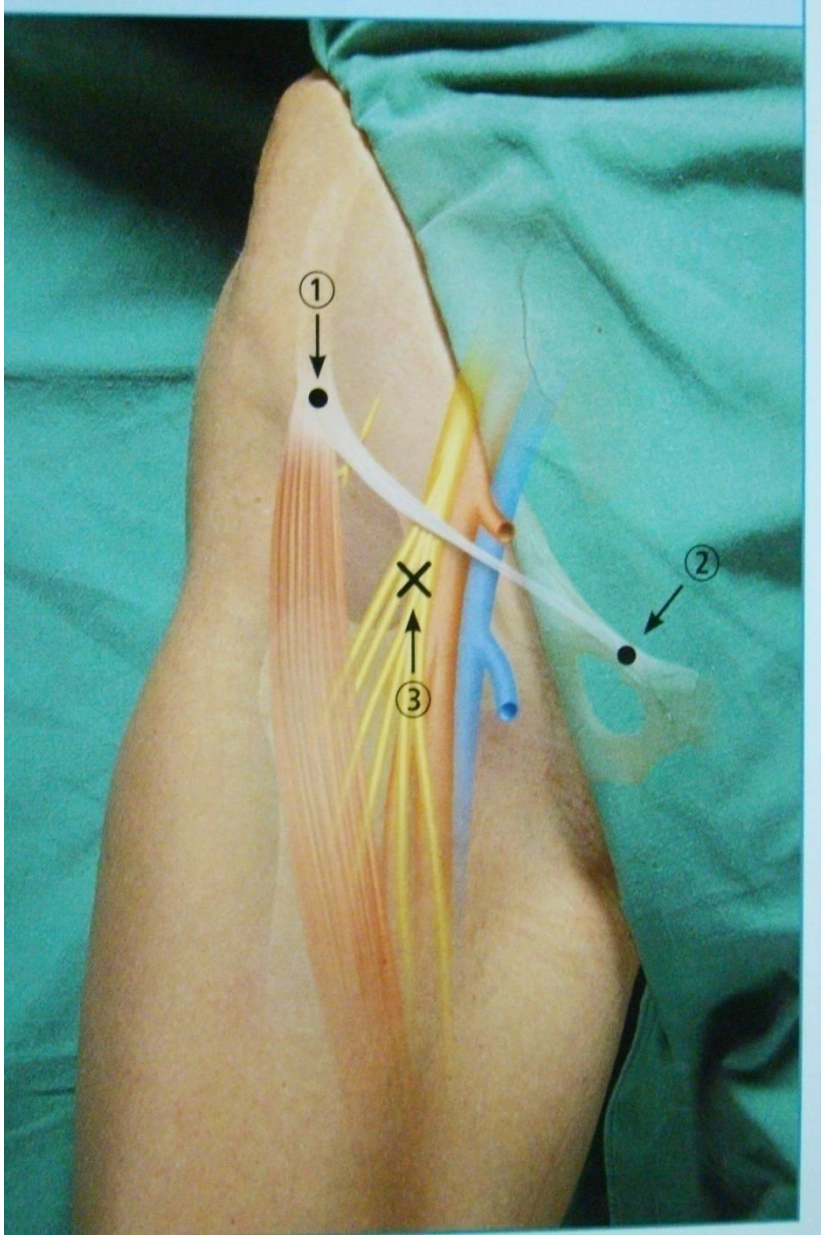
- Technique

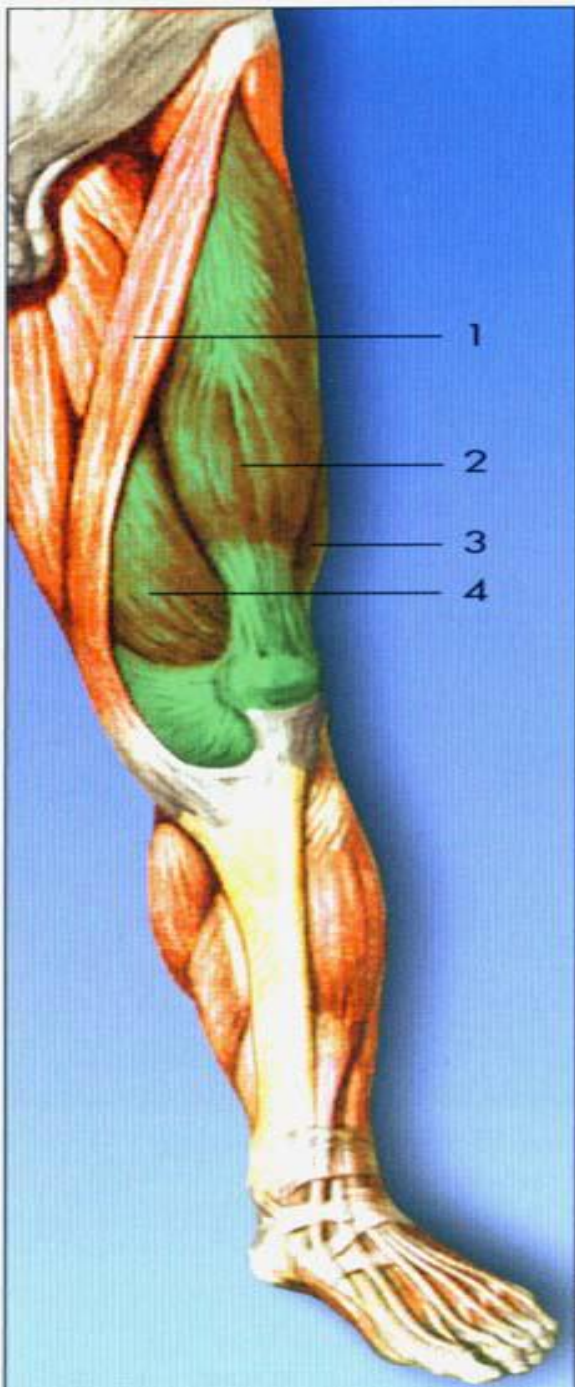
- At a point 1-2 cm below the inguinal ligament and 1-2 cm lateral to femoral artery
- Needle: 50 mm
- Direction: 45degree proximal/ perpendicular to the skin
- volume: 20 -30 ml of 0.25% bupivacaine
- Depth: 30-50 mm
 - two distinct “pops” may be felt (fascia lata, fascia illiacus)

Two distinct “pops”



Anatomical landmarks





Stimulation – Contraction of the quadriceps femoris muscle and patellar movements (“Dancing Patella”)

1. Sartorius muscle,
2. Rectus femoris muscle,
3. Vastus lateralis muscle,
4. Vastus medialis muscle

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2. Lateral cutaneous nerve of thigh block

- Indication

- Anaesthesia for lateral thigh (hip/femoral operations)

- Landmarks

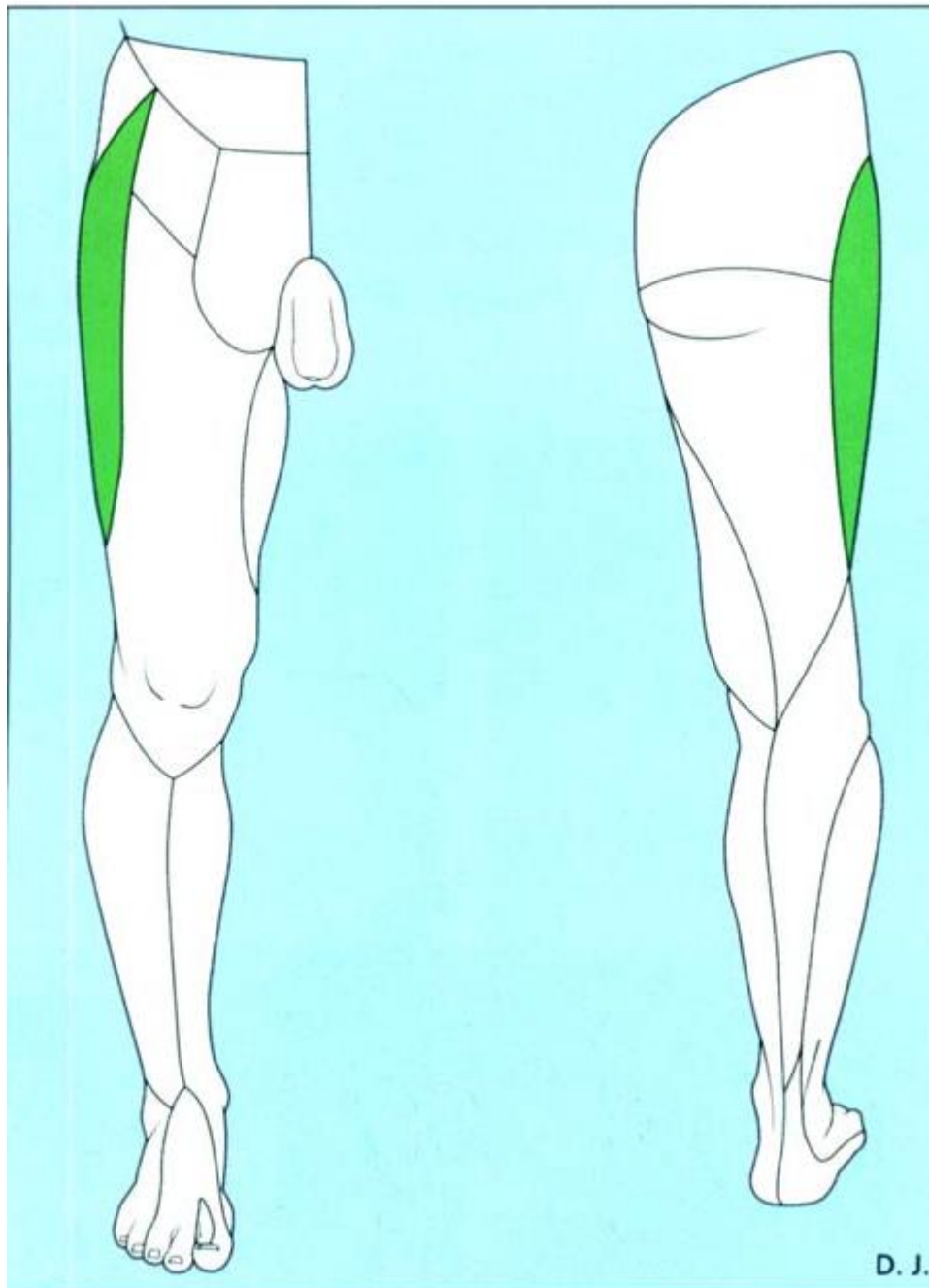
- ASIS, Inguinal ligament

- Technique

- At a point 2cms medial and 2cms inferior to the ASIS (below the inguinal ligament)
- 21-23G needle, perpendicular to the skin at a depth of 1-3cms feel for click as the fascia lata is pierced.
- Infiltrate above & below the fascia (total 10ml of LA)

- Complications —

- Accidental femoral NB

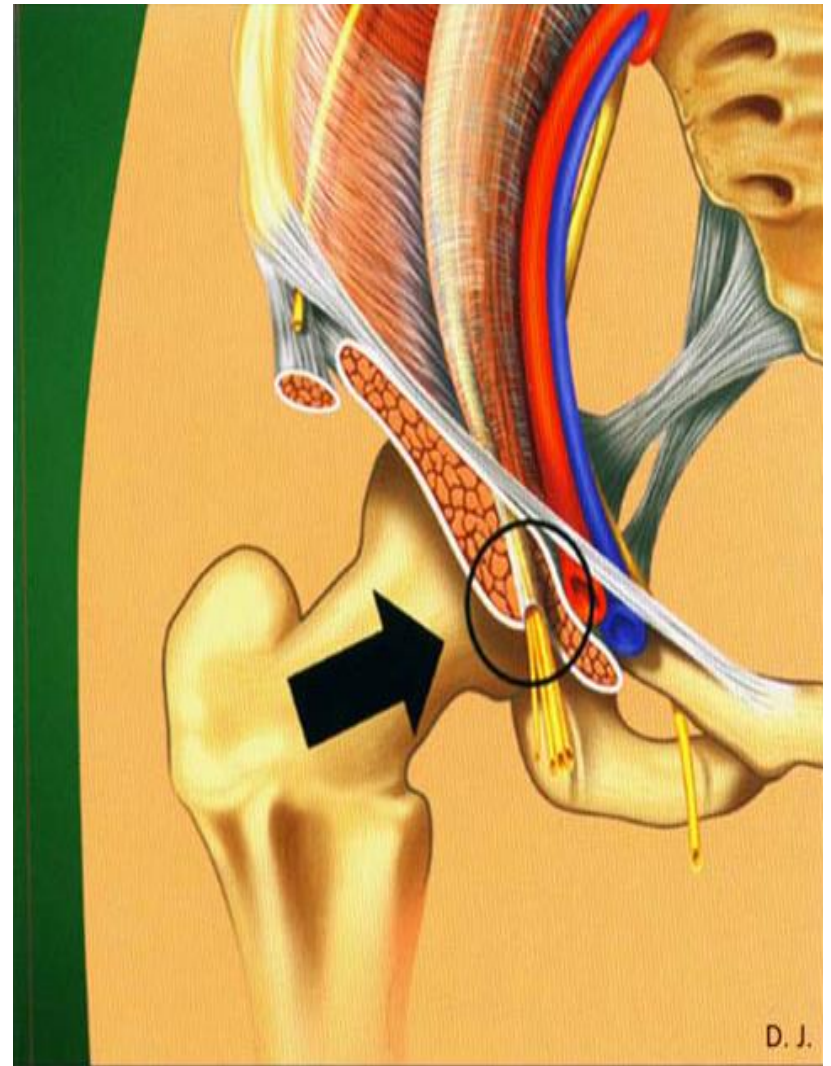
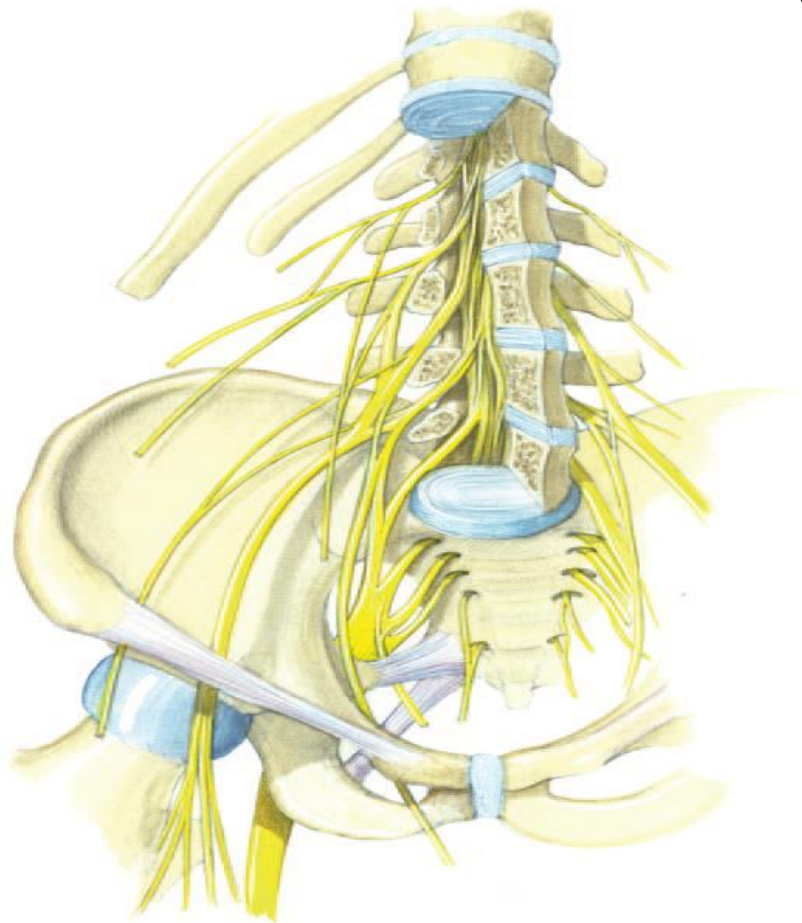


Cutaneous
innervation area of
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cutaneous nerve

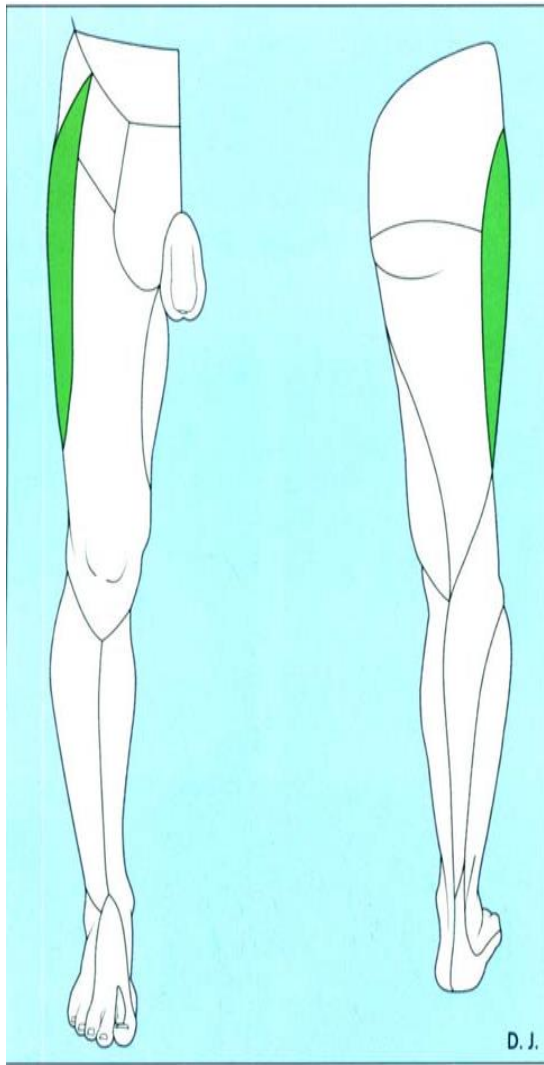
3. Three in one Block

- Definition

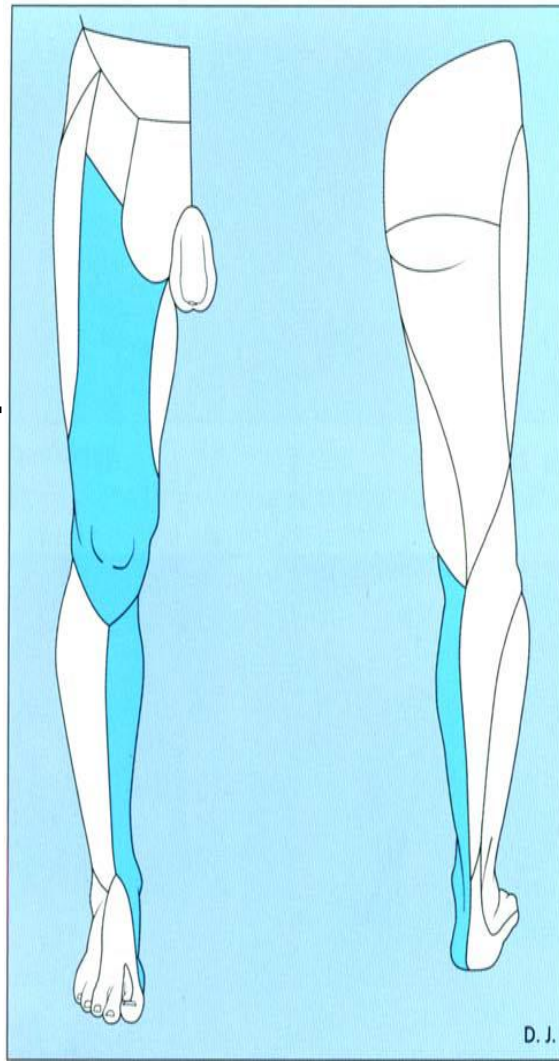
- The "three-in-one" block is an infero-anterior approach to the femoral nerve, lateral femoral cutaneous nerve and obturator nerve.
- These three nerves are blocked with a single injection into the common connective tissue and neural sheath immediately below the inguinal ligament.



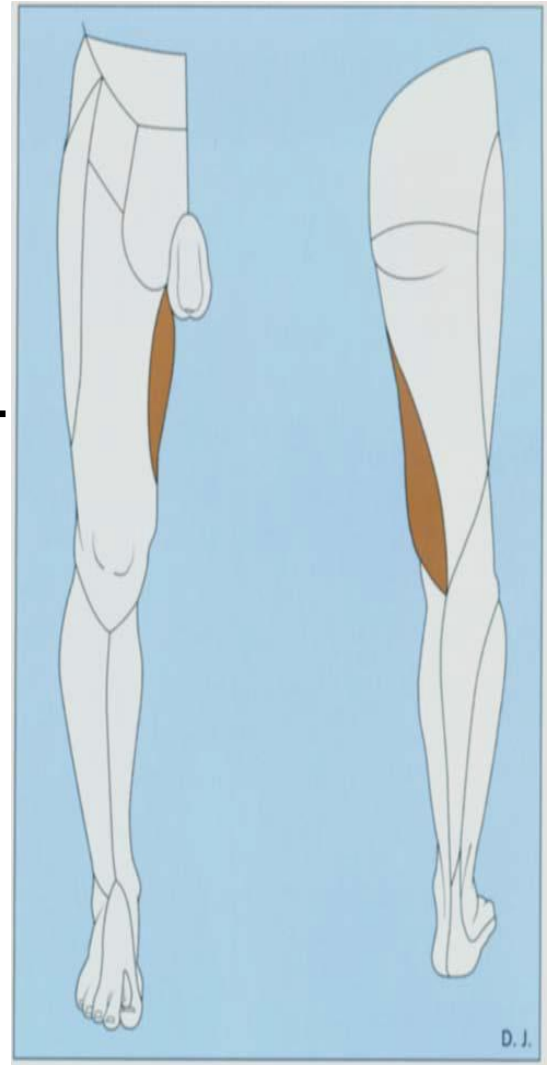
- Common connective tissue and neural sheath



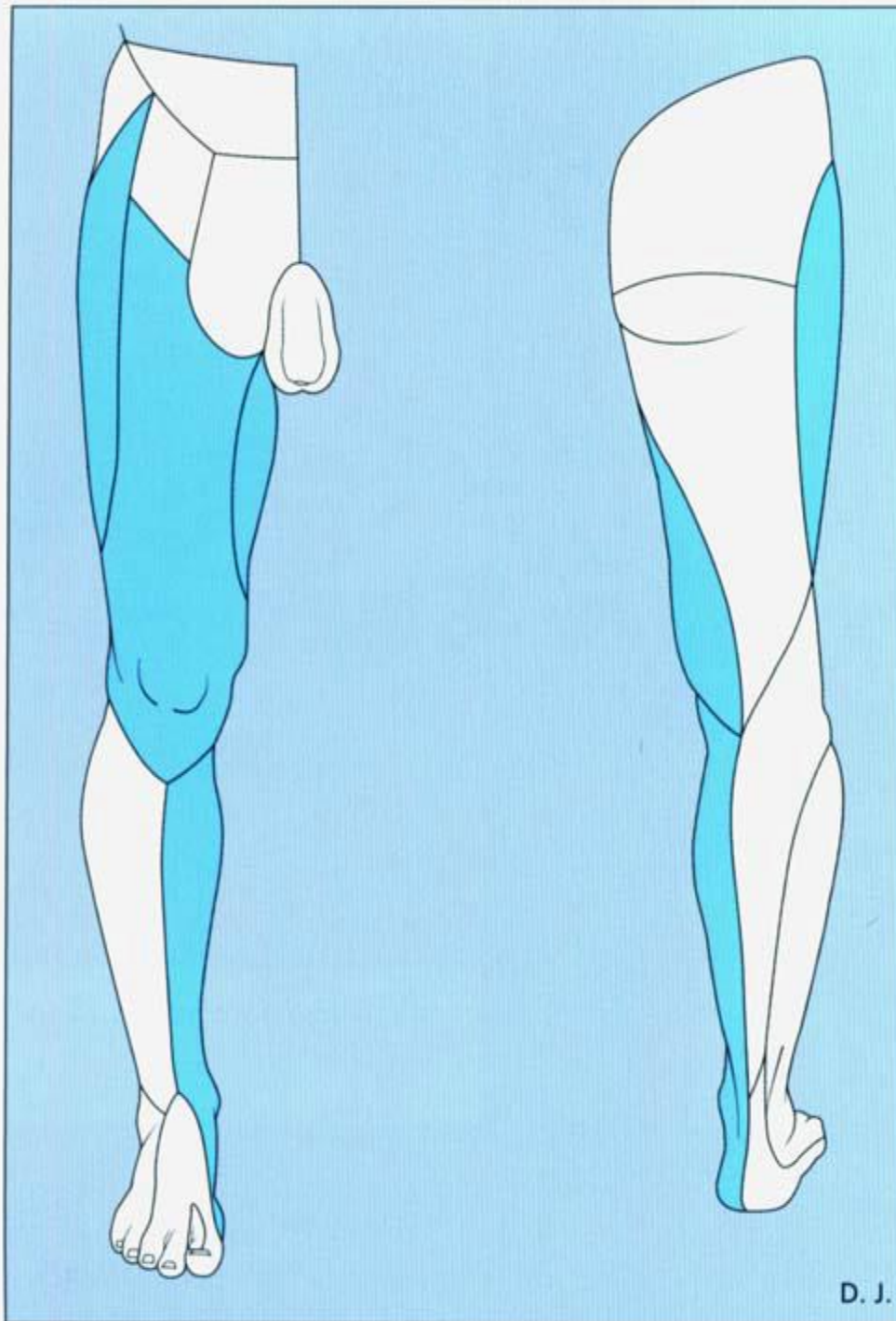
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- The neural areas most frequently blocked after administration of a "three-in-one" block

Technique

- Landmark and site of injection - same as FNB.
- Differences with FNB
 - a) Needle direction is 30-40° to the skin
 - b) Larger volume of LA ?
 - c) Compression & massage after injection and then flexing of the thigh for about 1 min.



Once you get a good twitch

- Do not advance the needle further!
- The stimulation current is reduced to 0.3 mA.
- Slight twitching suggests that the stimulation needle is in the immediate vicinity of the nerve.
- Aspiration test.
- Test dose of 3 mL local anesthetic
- During the injection, the twitching slowly disappears.
- Incremental injection of a local anesthetic (injection- aspiration after each 3-4 mL)

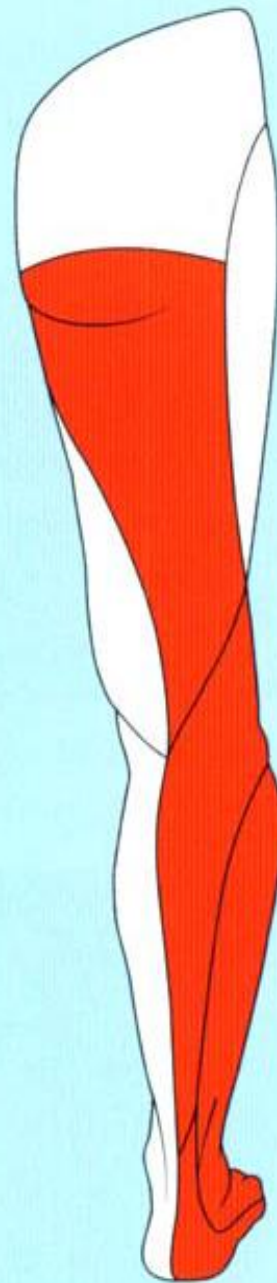
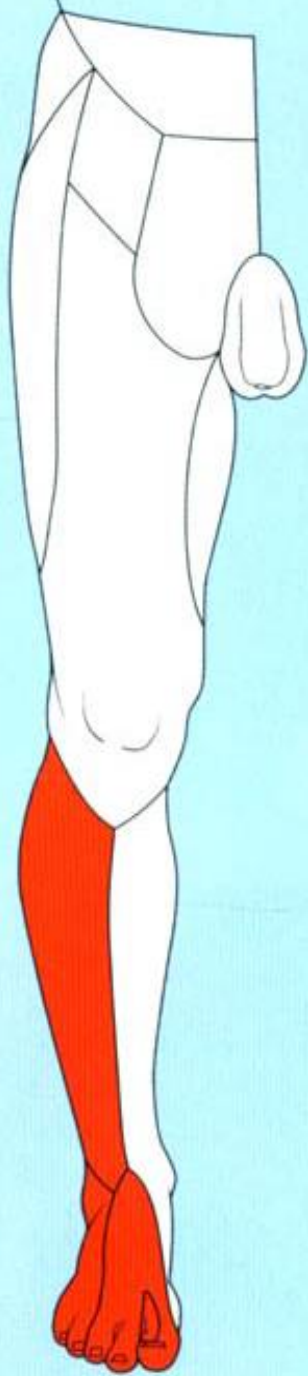
● Volume

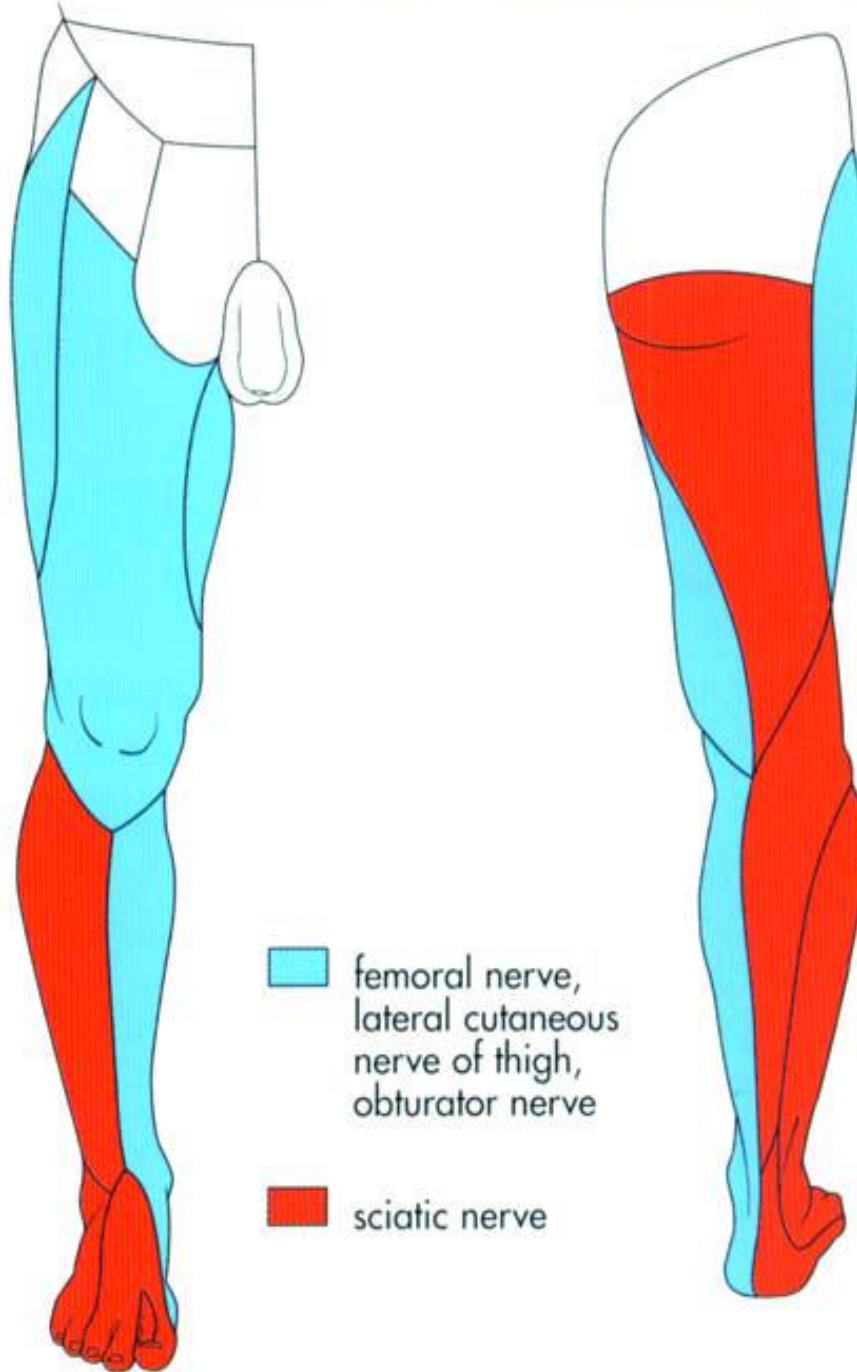
- 10 to 15 mls (Isolated femoral nerve)
- 20 to 30 mls (femoral and lateral cutaneous nerve of thigh) “2 in 1 block”
- 30 to 40 mls (femoral, lateral and obturator nerves) “3 in 1 block”

- Complications:

- Vascular puncture
- Inadvertent intravascular injection
- And other common PNB complications

4. Sciatic Nerve Block





Comparison of the innervation areas of the femoral nerve, lateral femoral cutaneous nerve and obturator nerve (blue) with the innervation area of the sciatic nerve (red)

Indication

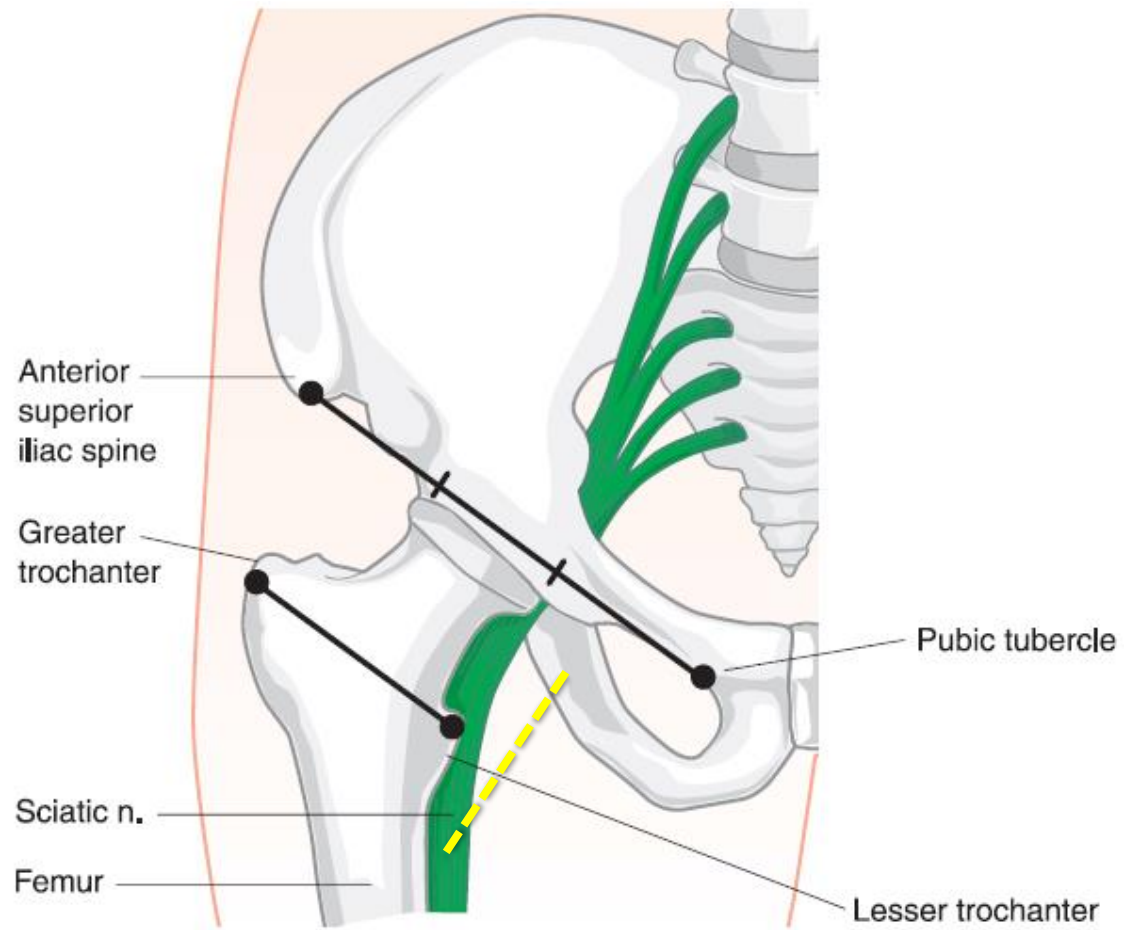
- Because of its wide sensory distribution, the sciatic nerve block can be used, together with a saphenous or femoral nerve block, for any surgical procedure below the knee.
- Solely – Ankle and Foot surgery,
 - Postoperative analgesia for lower limb amputation

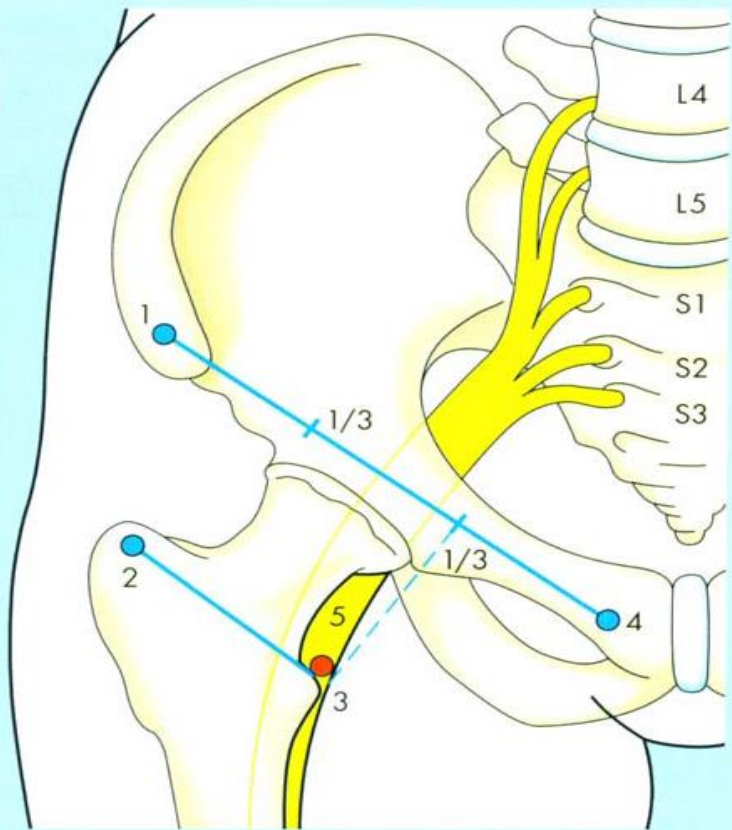
Landmarks: Anterior approach (**Beck**)

- Anterior superior illiac spine (ASIS), Pubic Tubercle (PT), Greater Trochanter (GT)

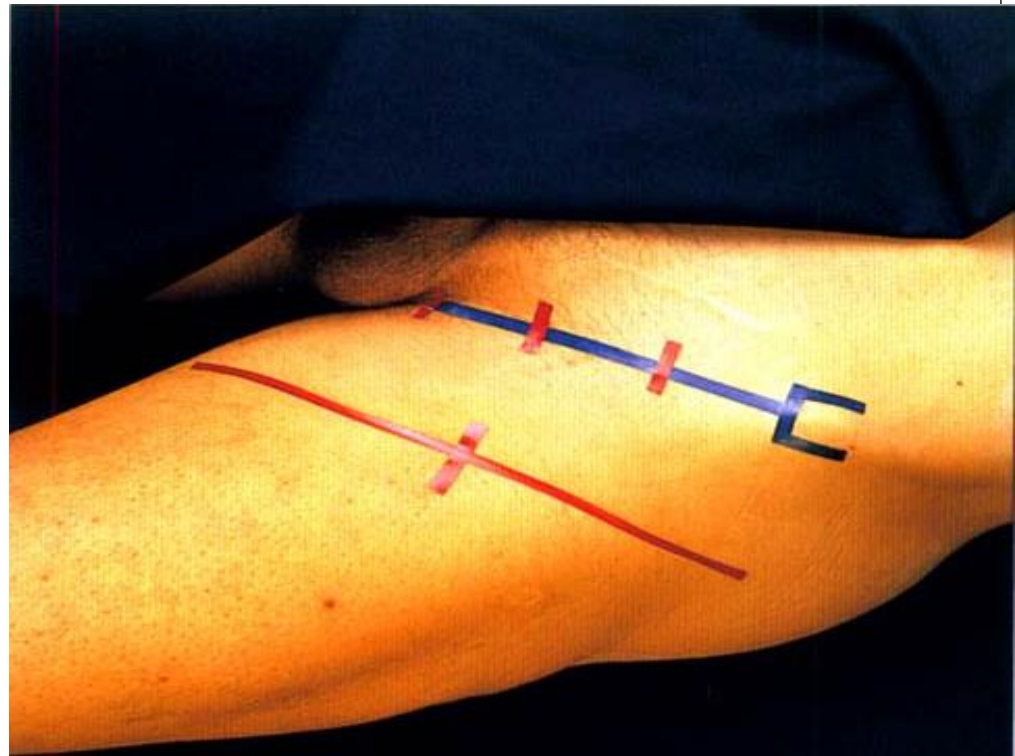
○ Technique

- Draw a line connecting the ASIS to the pubic tubercle
- Draw a further parallel line from the greater trochanter
- Drop a perpendicular line from the junction of middle and medial thirds
- Mark where the second and third line crosses





D. J.



D. J.

- Needle: 100 mm insulated
- Direction:
 - Needle is advanced directly posteriorly with slight lateral intent until it contacts bone (the lesser trochanter of the femur/medial aspect of femur)
 - The needle is redirected medially past the femur (a further 2-3cms),
- Depth: 8-10cms

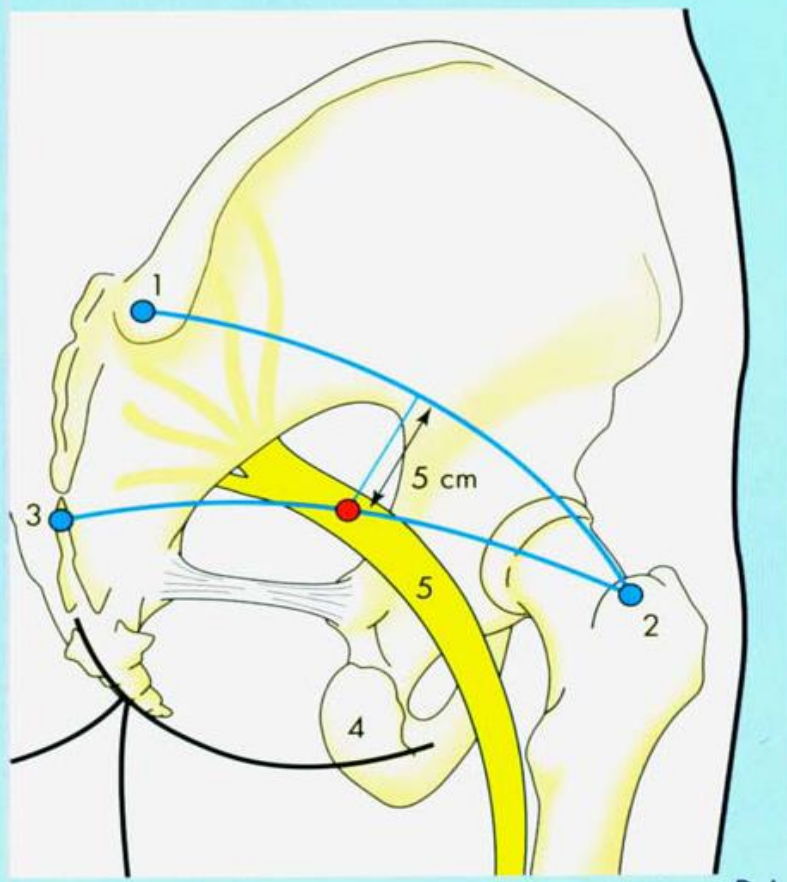
○ Landmarks Posterior approach (Labat)

- Posterior superior illiac spine (PSIS), Greater Trochanter (GT), Sacral Hiatus (SH)



○ Technique

- With patient in Sim's position, operative leg uppermost
- Draw a line connecting the PSIS to the GT, dropping a perpendicular from its mid point
- Mark where this crosses a line joining the sacral hiatus and GT.



D. J.



- Puncture depth

5 – 8 cm

- Positive stimulatory response

From the peroneal or tibial nerves:

Extensors or flexors of the foot or toes

- Dosage

20-30 ml of 0.25% Bupivacaine

- Avoid **adrenaline** containing solutions near sciatic nerve since blood supply may be damaged
- **Complications:** Femoral vessel damage, Accidental intravascular injection, Nerve damage

Which technique do you prefer ?

Anterior or posterior?

5. Popliteal Nerve Block (Lateral approach)

○ Indication

- Ankle and foot surgery

○ Landmarks

- Patella, biceps femoris, vatus lateralis

○ Technique

- Flex the knee
- Identify vatus lateralis above and tendon of biceps femoris below
- Mark the groove between, drop a line from the superior border of the patella
- Mark the point of intersection



○ Needle

- 50mm insulated

○ Direction

- 15° caudad, 30° posterior

○ Depth

- Common peroneal (fibular) – 10-20mm,
- Tibial 30-50mm

○ Stimulation

- Tibial – Plantar flexion of the foot
- Fibular – dorsiflexion/ eversion of foot

○ Volume – 10-15ml at each nerve

Clinical tips

- 75% of sciatic nerve divide within 10cm of popliteal crease.
- If fine movements of the needle cause both fibular and tibial nerves to be stimulated assume nerves are close enough together to use a single 20-30 ml injection of LA.
- Complications — same as FNB

6. Ankle and Foot nerve Blocks

- **Indications:** Forefoot and toe surgery
 - Deep and superficial fibular (peroneal) NB
 - Saphenous NB
 - Tibial NB
 - Sural NB
- **Patient position:**
 - The leg is kept in neutral position

- **Puncture site:**

- The **deep** and **superficial** peroneal and **saphenous** nerves can be blocked by using the same site.
 - First, a line is drawn between the two malleoli on the dorsal aspect of the foot
 - Second, the patient is asked to do an opposed extension of the foot to identify the **extensor hallucis longus**.
 - Third, the patient is asked to do an opposed extension of the big toe to identify the **extensor digitorum tendon** of the big toe.
- The needle is introduced between the two tendons at the level of the inter-malleoli line.

6.1 Deep Peroneal Nerve

- The needle is introduced perpendicularly to the skin until bone contact.
- At that point the needle is moved slightly and after negative aspiration 4-6 ml of LA is injected slowly.
- Next, the needle is repositioned at the level of the skin.



6.2 Superficial Peroneal Nerve

- 4-6 ml of LA is injected subcutaneously and laterally up to the lateral malleolus.
- Next, the needle is repositioned at the level of the skin.



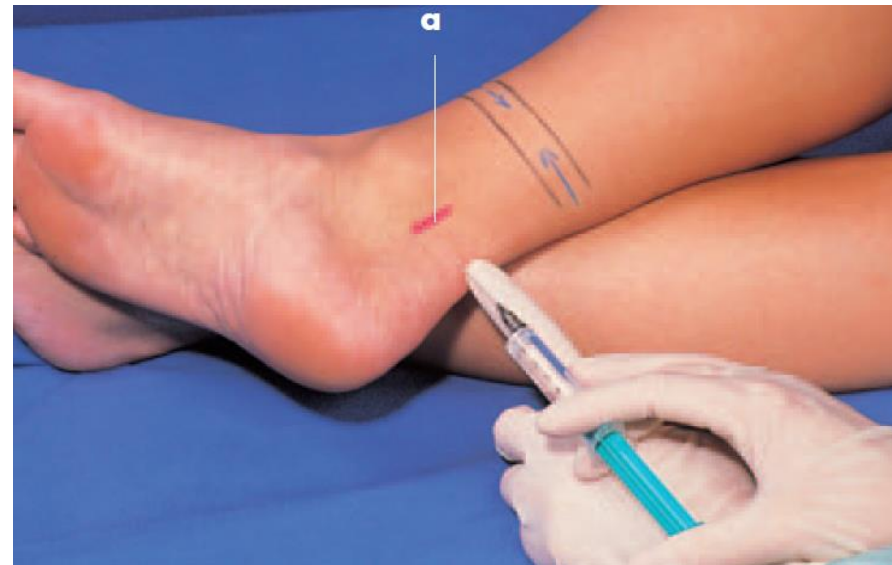
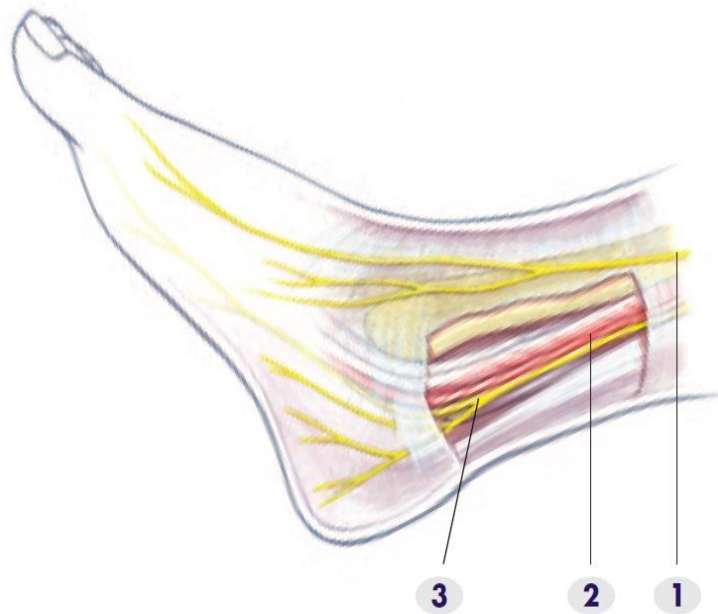
6.3 Saphenous Nerve Block

- 4 ml of LA is injected subcutaneously and medially up to the medial malleolus.
- The block of the superficial peroneal and saphenous nerves resembles a ring between the malleoli.



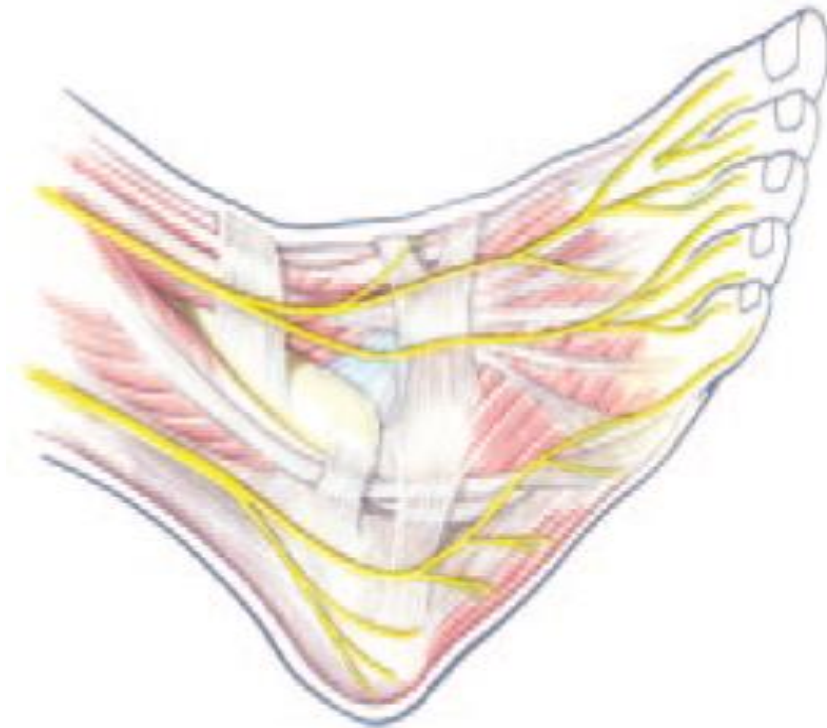
6.4 Tibial Nerve Block

- Draw a line joining the **medial malleolus** to the posterior inferior border of the **calcaneum**
- Mark a point just posterior to the posterior **tibial pulse** (half way)
- Needle – 25-50mm
- Stimulation – Plantar flexion toes, paresthesia to the sole of foot/toes
- Volume 5-8ml



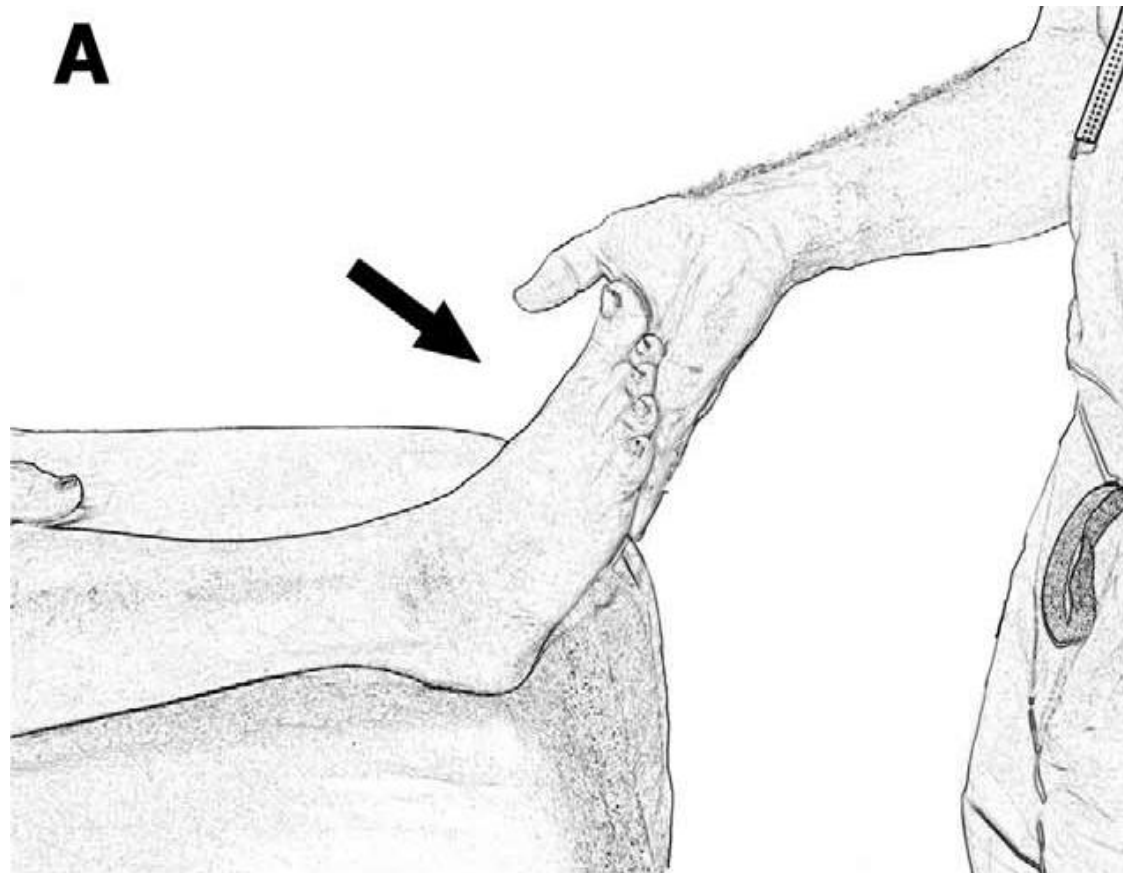
6.5 Sural Nerve Block

- Infiltrate subcutaneously from lateral malleolus to lateral border of Achilles tendon with 5ml of LA.
- Painful block !!!



Assessment of lower-extremity nerve
block by the **Four P's**

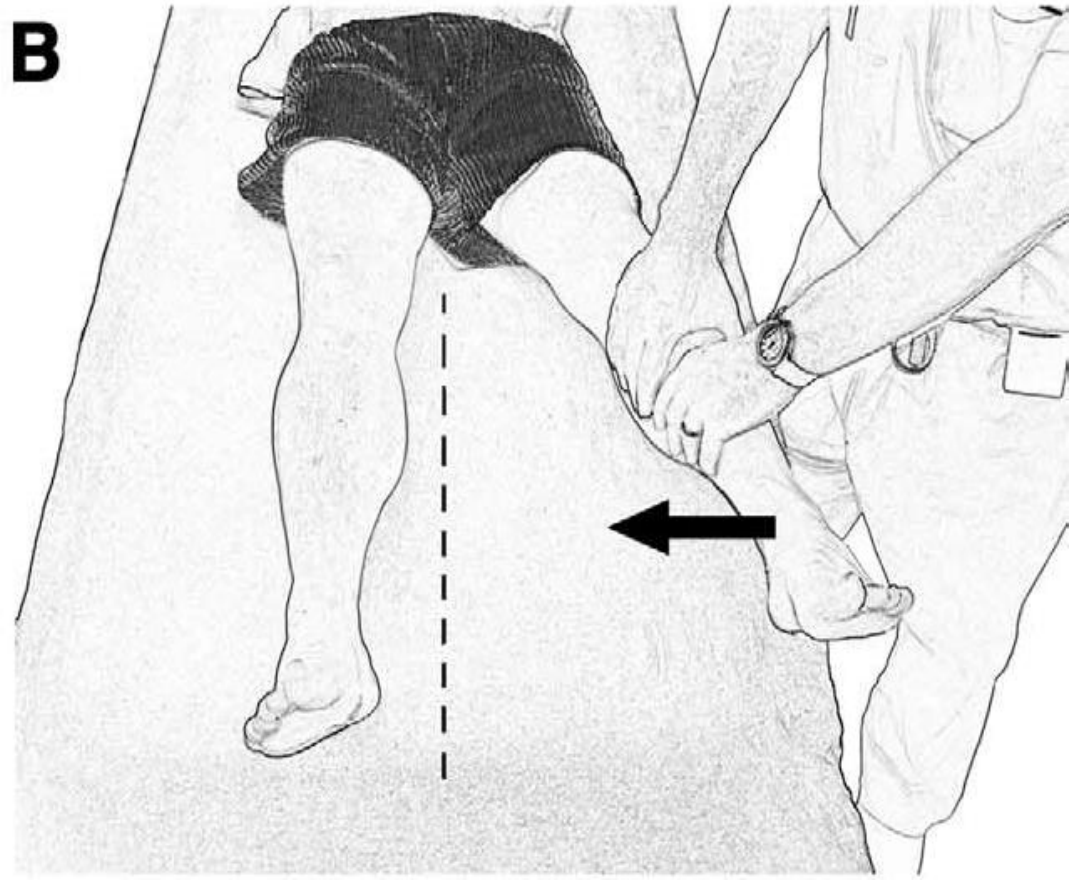
A. **Push;** Inability to plantar flex the foot against resistance indicates:
sciatic nerve blockade.



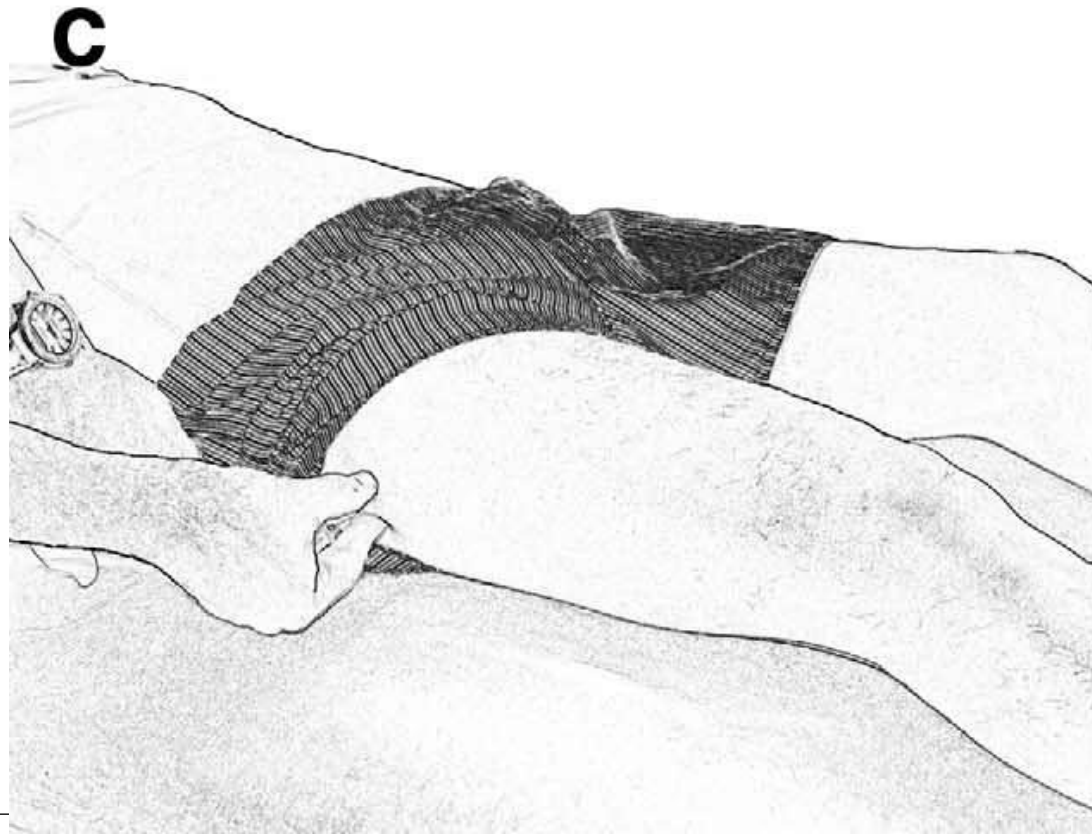
B. Pull: Resist the patient's attempt to adduct the leg toward the midline.

Weakness signals
conduction block of:

the obturator nerve.



C. **Pinch**: inability to detect a pinch on the proximal lateral thigh → shows anesthesia within the *lateral femoral cutaneous nerve* distribution.



D. Punt; Raise the knee and asks the patient to extend the knee against the knee against resistance. Inability to perform this task signals *successful femoral nerve block*.

D



References

- American Society of Regional Anaesthesia and Pain Medicine, 2002 journal.
- Gray's Anatomy
- Regional nerve blocks and infiltration therapy
- Pocket compendium of peripheral nerve blocks

10 Q!!!